

Competing Complements in Public-Private Hospital Markets

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Abstract

Public hospitals in many countries provide subsidized care that competes with private hospitals, while training physicians who later join the private sector. I study this tension using the staggered construction of public hospitals in Malaysia from 1996 to 2013. A new public hospital increases private hospital entrants by 36 percent. This crowd-in requires both labor-supply spillovers from public hospitals and districts large enough to support private entry. Reallocating public hospitals to larger districts would raise crowded-in private capacity by roughly a third but reduce equity in access. These findings show public hospitals trade off equity-driven allocation against total healthcare capacity.

JEL Codes: I11, H44, I18, J44, L33, O15

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1 Introduction

Whether government spending crowds out or crowds in private investment is a classic question in economics. This question is important for many markets where publicly owned providers coexist alongside private for-profit firms. The existing literature shows that offering a public option reduces private health insurance coverage (Cutler and Gruber, 1996; Gruber and Simon, 2008), affects private school enrollment (Dinerstein and Smith, 2021) and reduces demand for private investment in municipal broadband markets (Wilson, 2025). On the other hand, investing in the public sector can raise private school quality through competition (Andrabi et al., 2024), and public research labs can generate local R&D spillovers to private firms through researcher mobility (Bergeaud et al., 2025). These opposing effects reflect a tension between competition and complementarities, as public provision may reduce demand for private alternatives, but it may also generate benefits that complement private firms.

Understanding this tension is especially important for healthcare in many countries and especially in developing contexts, where the private sector delivers a large and growing share of healthcare services (WHO, 2020). The conventional rationale for public spending is equity driven—serving underserved areas where private markets do not enter. Yet, if the public sector also produces inputs that complement private hospitals, the same public investment can simultaneously expand direct provision and crowd in private supply. Understanding when crowd-in or crowd-out occurs is important for policymakers to decide how to allocate scarce public hospital construction budgets.

In this paper, I study this tension between competition and complementarities using staggered public hospital construction in Malaysia between 1996 to 2013¹. During this period, Malaysia underwent a rapid expansion of public hospitals in response to backlash against proposed healthcare privatization. I construct a new district-year panel dataset linking administrative records on public hospital construction to private hospital entry from survey data.

A simple entry model framework with public and private hospitals predicts that crowd-in or crowd-out depends on whether public hospitals generate complementarities that exceed competition. Complementarities can arise through demand or supply channels. On the demand side, public hospitals may create demand spillovers through referrals or congestion at existing facilities. On the supply side, public hospitals—holding a monopoly on the training of healthcare professionals—may generate labor spillovers that reduce private variable costs. Conversely, competition may arise for both patients and healthcare professionals, as public hospitals may draw patients away from private hospitals and reduce existing labor supply for private hire.

Using a staggered event study design, I find that a new public hospital crowds in an additional 0.465 private hospitals in treated districts, against a pre-treatment mean of 1.440. This main effect conceals substantial heterogeneity by whether a public hospital provides specialty medical services or is a general non-specialist public hospital. Specialist public hospitals increase the stock of private hospitals in treated districts by approximately 0.785 against a pre-treatment mean of 2.571. Non-specialist hospitals in contrast, show null effects. The crowd-in effect concentrates in larger districts.

¹Malaysia experienced a rapid growth in its hospital industry between 1980 and 2020. In contrast, most countries now experience hospital closures (Gaynor et al., 2012; Alexander and Richards, 2023; Ghislandi et al., 2025).

Several features of the event study design support a causal interpretation. Treated and control districts follow parallel pre-treatment trends in private hospital entry, are balanced on key pre-treatment characteristics, and show small differences between predicted private entry and public hospital construction once district fixed effects are included. The specialist result is directionally consistent across four alternative staggered difference-in-differences estimators, synthetic difference-in-differences, and matching.

The heterogeneity by hospital type suggests one complementarity in particular: the training of specialist physicians. Specialist public hospitals house specialist residency programs. Non-specialist hospitals train nurses and general practitioners but not specialists. This distinction matters, given that a specialist takes years of postgraduate residency under attending supervision to produce, and is the key input for private hospitals². Other mechanisms, such as agglomeration externalities, patient referral networks, or localized infrastructure improvements, could in principle generate complementarities as well. However, three additional pieces of evidence support the specialist physician spillovers interpretation.

First, census records show that the district-level stock of private specialist physicians increases after specialist public hospital construction, with timing consistent with specialist residency duration. This occurs despite public hospitals simultaneously reducing private hospital admissions upon entry. The second piece of evidence comes from a falsification test that holds specialist public hospital training infrastructure constant and varies only whether new bed capacity was added. The effects of these bed expansions on private hospital entry are null, suggesting that the crowd-in effect is not driven by capacity expansion.

²Private hospitals are regulated by the 'Private Healthcare Facilities And Services Act 1998', and the act requires a specialist physician to be among the founding staff of a new hospital.

The third comes from more recent data on individual-level specialist physicians. A cross-sectional snapshot in 2025 of 12,503 registered specialists shows that approximately 40 percent of specialists whose compulsory service bonds have expired practice in the private sector. The public sector trains all specialists through its monopoly on clinical residency, yet the private sector, which accounts for only 28 percent of total hospital beds, absorbs 40 percent of the specialist supply. Among 244 specialists from recent cohorts observed transitioning from public to private practice within the registry window, 62 percent remain in the same district as their last public posting or within the greater *Kuala Lumpur* metropolitan area. The pipeline from public training to private practice is thus large, geographically concentrated, and consistent with the timing and location pattern that specialist physician spillovers predict.

The spatial pattern of private entry within treated districts also highlights the tension between competition and complementarities. Within treated districts, private entry response is non-monotonic in distance from the new public hospital: lowest in the immediate vicinity where patient competition is most intense and peaking at intermediate distances where the physician labor pool remains accessible. The non-monotonic gradient provides within-district evidence that the district-level crowd-in reflects the tension between opposing forces, rather than an artifact of cross-district heterogeneity.

Using the reduced form estimates, I calculate a back-of-envelope value of a specialist public hospital to private entrants. The key assumption is that private hospitals' variable profit scales with district population, so that private entry response to market size identifies the entry response to profit. The valuation nets two opposing channels. The new public hospital competes for patients: incumbent private hospitals lose 37 to 46 percent of baseline patient volume. Private entry

increases regardless, and the entry response puts the net gain to a marginal private entrant at 12 to 31 percent of its baseline variable profit. The labor-cost saving from publicly trained specialists must therefore cover both the demand loss and the net gain, amounting to RM 16 to 22 million per entrant. This is roughly half the cost of building a private hospital.

Reallocating the specialist public hospitals placed in districts with no private hospitals at baseline toward districts with high private entry potential would crowd in roughly a third more private capacity: 193 additional private beds, 38 percent of the bed capacity of an average new specialist public hospital, at no additional public expenditure.

If the social planner's objective is to maximize total healthcare capacity, these results suggest a type-specific allocation strategy. Non-specialist hospitals should be allocated towards districts where the private sector will not enter, while specialist hospitals can be targeted at marginal districts with high private entry probability. But this strategy also highlights an equity-efficiency tradeoff for hospital allocation policy. Allocations that maximize equity deliver minimal private crowd-in, while targeting districts to induce more private entry delivers less direct equity benefit. With limited fiscal space across many countries, this tradeoff becomes particularly salient for policymakers deciding how to allocate scarce public construction resources.

Related Literature. This paper contributes to three literatures. The first studies public provision and private market responses across education (Epplé and Romano, 1998; Hoxby, 2000; Dinerstein and Smith, 2021), health insurance (Cutler and Gruber, 1996; Duggan and Scott Morton, 2006; Saltzman, 2023), pharmacies (Atal et al., 2024), broadband (Wilson, 2025), and consumer goods (Jiménez Hernández and Seira, 2022). Recent work shows that crowd-in is possible under specific condi-

tions: competitive pressure from public school funding raises private school quality (Andrabi et al., 2024), and public research labs generate local R&D spillovers to private firms partly through researcher mobility (Bergeaud et al., 2025). A large hospital and insurer competition literature exists for the United States (Gaynor et al., 2014; Ho and Lee, 2017; Shepard, 2022), but few studies examine public-private hospital competition in developing countries, where patients pay out-of-pocket and public hospitals dominate specialist training (Das et al., 2008; Banerjee et al., 2024). My contribution is to document that public provision crowds in or crowds out private investment depending on whether it expands an input that private firms also require, in a setting where the public sector controls that input.

The second literature is mixed public-private competition, which has been largely theoretical (Cremer et al., 1991; De Fraja and Valbonesi, 2009; Klumpp and Su, 2019) with limited empirical work on hospitals (Bloom et al., 2015; Herr, 2011). I show that competitive outcomes in mixed markets depend on the specific characteristics of public providers, particularly whether they generate complementarities that benefit private competitors.

The third literature is place-based policy (Kline and Moretti, 2014; Juhász et al., 2024). I add that public provision can stimulate private investment through workforce training, a channel distinct from targeted industrial policy.

The paper proceeds as follows. [Section 2](#) provides context and data on the Malaysian public and private hospital industry. [Section 3](#) presents descriptive facts about the public-private hospital market. [Section 4](#) sets out a conceptual framework for the competition-complementarity tension. [Section 5](#) presents the empirical findings. [Section 6](#) concludes.

2 Context and Data

2.1 History of the Public-Private Malaysian Health System

Following Malaysia's independence from the British Empire in 1957, the Ministry of Health established a network of public hospitals to ensure universal access to healthcare services. This public system operated as the dominant healthcare provider until the 1980s, when Malaysia adopted a series of broader economic liberalization policies extended to the healthcare sector. The privatization wave of the early 1980s was a key policy shift in Malaysia's health system. The government actively encouraged private investment through tax breaks for medical devices and private health insurance (Barracough, 2000). These nationwide policies allowed private investors to enter markets based on profit considerations rather than central planning directives.

However, by the mid-1990s, political resistance to healthcare privatization emerged as a constraint on further market oriented policies. The government's initial plans to corporatize public healthcare services faced significant opposition from the ruling coalition's constituents, who viewed potential reductions in subsidized public healthcare as a threat to equitable medical care.³ This political backlash resulted in a policy reversal that reinforced the government's commitment to maintaining a robust public healthcare system alongside the growing private sector.

The 1995 general election is an important moment that shaped Malaysia's public-private healthcare system. Rather than pursuing further privatization of public

³In 1985, Prime Minister Mahathir Mohamad announced a series of privatization and corporatization policies across multiple industries. This led to political concern from the main coalition's constituents, as the government contemplated reducing subsidized public healthcare services. In the 1995 Malaysia general elections, the government retracted all policies related to corporatizing public healthcare services and increased the number of public hospitals to show commitment to retaining a public-dominant health system (Barracough, 1997, 2000).

services, the government responded by expanding public hospital capacity and reaffirming subsidized public healthcare as a core public good. As a result, Malaysia developed hospital markets where private hospitals operate as an expensive alternative to a heavily subsidized public system, rather than as replacements for public provision.⁴

Importantly for the event studies, the government's renewed commitment to public hospital expansion post-1995 provides an empirical opportunity to examine the effects of public hospitals on private investment. The construction of public hospitals following the electoral mandate creates variation in public hospital entry location and timing, which I use in the event study analyses to identify causal effects on private hospital entry.

2.2 Hospital Classification, Regulation, and Pricing

The Ministry of Health classifies public hospitals into four tiers based on specialization and capacity. State hospitals and major specialist hospitals (Level 4) offer the full range of specialty and subspecialty services. Minor specialist hospitals (Level 3) provide a more limited set of specialist services. Non-specialist hospitals (Levels 1–2) provide general medical and nursing care staffed by medical officers rather than specialists. In the event study analysis, I group state hospitals, major specialist hospitals, and minor specialist hospitals as “specialist” hospitals, since all three tiers operate residency training programs. Non-specialist hospitals do not train specialists.

⁴The 1997–98 Asian Financial Crisis caused Malaysia's GDP to contract by 7.4 percent in 1998. Public expenditure fell short of planned targets in 1998 and 1999, likely delaying some hospital construction projects planned under the Seventh Malaysia Plan (1996–2000). Because the crisis affected all of Malaysia simultaneously, these delays are absorbed by year fixed effects in the event study specification.

Public hospital allocation follows a multi-tiered process embedded within Malaysia's five-year development planning cycle. Hospital funding is allocated to districts based on the Malaysia Plans, which are national development blueprints that prioritize healthcare accessibility and population coverage. The Ministry of Health collaborates with State Economic Planning Units to identify districts requiring new healthcare infrastructure based on demographic projections, existing facility capacity, and accessibility gaps (see [Figure A.7](#) for excerpts from official planning documents). After districts receive funding allocations through this centralized planning process, local health authorities work with district officials to identify specific sites that maximize population access while considering factors such as land availability, transportation networks, and proximity to existing health facilities.

In contrast, private hospital entry operates under a regulatory framework established by the Private Healthcare Facilities and Services Act 1998. While the Ministry of Health retains approval authority for private hospital licenses, the regulatory standard primarily requires demonstration of sufficient local demand rather than adherence to national planning objectives. Private hospitals can choose any location within a district based on commercial considerations such as population density, income levels, and competitive positioning. In short, private entrants seek to make profits, while public hospitals are centrally allocated based on accessibility objectives.

The regulatory framework governing pricing differs between sectors. Public hospitals operate under a unified national pricing structure, with the government setting standardized fees for all services across the country but varying by room types. These prices are heavily subsidized. For example, a normal vaginal delivery in a public hospital costs Malaysian citizens MYR 100 (approximately USD 24) in a

third-class ward while private hospitals charge a mean of MYR 3,306 for the same service. Private hospitals face a more complex regulatory environment. While the 1998 Act establishes fee schedules for physician consultations and medical procedures, it does not regulate hospital-specific charges such as room fees, meals, and ancillary services. This partial price regulation allows private hospitals significant flexibility to price their services based on local market conditions and competition.

2.3 Physician Training and the Public-to-Private Pipeline

Malaysia requires all specialist physicians to complete a four-year mandatory service obligation in public hospitals. After a physician has obtained her basic medical degree, specialist training imposes additional years of public service. All physicians seeking specialization must complete a Master of Medicine (MMed) residency program lasting four to six years depending on the discipline, conducted in Ministry of Health hospitals.⁵

Trainees receive full sponsorship under the *Hadiah Latihan Persekutuan* (HLP) scholarship administered by the Public Service Department, which bonds them to public service for an additional four to seven years after completing specialization. Graduates can choose to break this bond by repaying the bond penalty and leave the public sector, or choose to enter private practice once their bond obligation is fulfilled.

Specialist posting to public hospitals after MMed graduation is centrally administered by the Ministry of Health's Medical Development Division based on identified workforce needs and post availability at specific hospitals, weighted by length of prior service and prior posting in district or rural hospitals (Ismail et al.,

⁵While specialist degrees are conferred by both public and private universities, clinical residency training takes place in Ministry of Health hospitals regardless of the degree-granting institution. A small number of specialists obtain qualifications abroad and register through the National Specialist Register upon return.

2025). Bonded specialists under HLP cannot decline assigned posts without buying out the bond. When a new specialist hospital opens, MOH allocates new specialists there based on the hospital's identified workforce needs. Over time, these posts fill from successive MMed cohorts, producing a localized concentration of trained specialists.

During the study period (1996–2013), formal dual practice in both public and private sectors was not permitted. The Ministry of Health introduced a limited dual practice policy in 2007 that allowed public specialists to treat private patients within the same public hospital for additional fees, but this arrangement does not extend to practicing at external private facilities. The primary channel through which publicly trained specialists enter the private sector is therefore full resignation from government service after completing bond obligations, rather than gradual moonlighting.

2.4 Data

I provide a brief overview of the data used in the event study and mechanism analyses, with further details in [Appendix A](#). A summary of the key variables is tabulated in [Table A.1](#). The event study data comes from a combination of administrative data and surveys conducted by the Ministry of Health. I estimate my event studies using a district-level panel across all Malaysian districts over 1996-2013.⁶ The analysis focuses on 25 new public hospitals (14 specialist and 11 non-specialist) that began operations between 1996 and 2013 across 25 treated districts, using the first treatment event in each district, and their impact on the stock of private hospitals.

⁶The sample includes districts in both Peninsular Malaysia and East Malaysia (Sabah and Sarawak). Eight of the 25 treated districts are in East Malaysia, where healthcare markets are more rural and private hospitals are less prevalent.

I construct this panel using data from the National Healthcare Establishment and Workforce Survey (NHEWS), which contains information about every hospital providing hospitalization services that was operational in 2013. Using each hospital's construction and opening dates, I backfill the count of public and private hospitals operating in each district for every year from 1996 to 2013. This implies that exits are not accounted for, though Malaysia's private hospital sector was in a period of rapid expansion during my study period (Barraclough, 2000). The final dataset includes all general and specialized hospitals providing acute curative care from both public and private sectors. I exclude specialized institutions (prison, defense, and education ministry hospitals) and long-term care facilities (rehabilitative and palliative care hospitals, nursing homes, leprosy centers, and psychiatric institutions).

To understand the mechanisms driving private hospital entry patterns, I use two additional outcome data measured at the district level. Data on private specialist physicians comes from the Population and Housing Census for 1970, 1980 and 1991⁷. I use district-level counts of self-employed physicians as a proxy for private specialist physicians, since all public specialist physicians are civil servants receiving wages rather than operating independently. This implies that my outcome combines private specialists alongside other private sector physicians such as general practitioners.

Data on private hospital utilization comes from the National Health and Morbidity Survey conducted in 1996, 2006, and 2011. This nationally representative survey interviews approximately 59,000 respondents in 1996 and 2006, and 29,000 in 2011. The survey asks about healthcare utilization in the previous year, including

⁷The 2000 census adopted revised ISCO occupation codes that combine physicians, veterinarians, dentists, and other medical professionals into a single category, precluding identification of specialist physicians. I therefore cannot construct district-level physician counts during the study period itself using census data.

private inpatient admissions. The survey weights allow for district-level estimation of private hospital utilization rates.

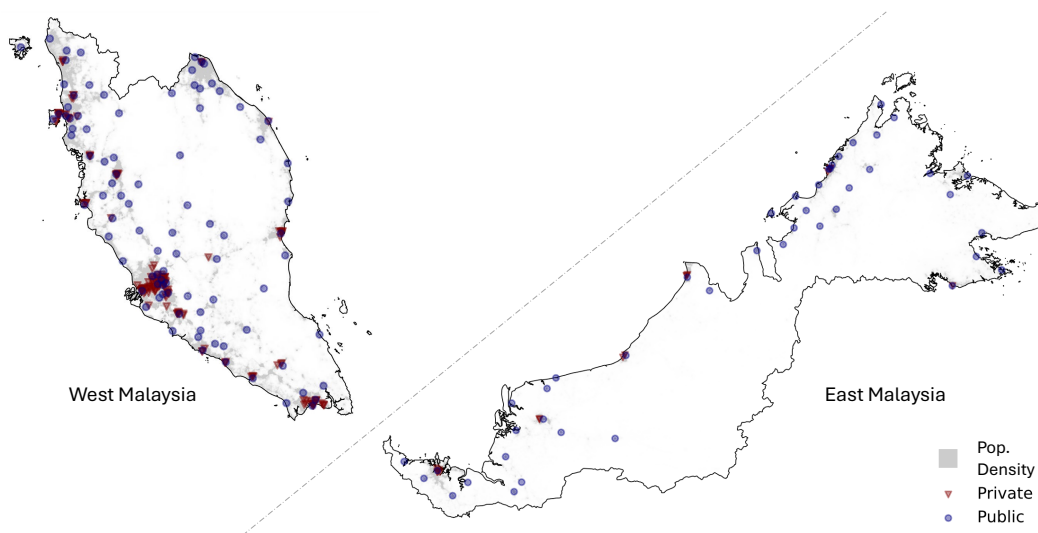
Data on individual specialist physicians comes from Malaysia’s Annual Practicing Certificate (APC) registry. I collect the complete registry of practicing specialists and cross-match it with the National Specialist Register (NSR) to identify specialist qualification years and fields of practice. The resulting dataset covers 12,503 registered specialists observed across 2023–2026, recording each physician’s primary practice location, sector (public, private, or university), specialty, medical school, and year of specialist qualification. The qualification year ranges from 1950 to 2024, providing variation in how long each specialist has had to complete bond obligations and potentially transition to private practice. Because the APC is a mandatory annual license required for all practicing physicians, the data captures the near-universe of actively practicing specialists in Malaysia regardless of sector. I use this data to describe the public-to-private transition rate by qualification cohort and the geographic concentration of transitions, providing direct evidence on the physician training pipeline described in Section [2.3](#).

Finally, I use ‘Health Facts’, an annual publicly available dataset containing hospital-level information on total beds from 2003–2012, to construct an alternative treatment of hospital upgrades. Health Facts covers the same hospitals as NHEWS, allowing me to identify existing hospitals that received significant capacity upgrades (defined as increases in bed count). I observe 35 such upgrades across different hospitals during this period. This alternative treatment tests whether the private hospital entry effects are specific to entirely new public hospital construction, or also occur when existing public hospitals expand their capacity.

3 Descriptive Evidence

Private hospitals locate in urban areas while public hospitals distribute more evenly across the country. [Figure 1](#) maps hospital locations in 2013 against population density. [Figure A.3](#) shows how this geographic pattern has changed over time. Between 1980 and 2013, private hospitals expanded primarily in urban centers while public hospitals grew in rural areas.

Figure 1: Public and Private Hospital Locations in 2013



Note: Hospital location data are from the National Healthcare Establishment and Workforce Survey (2013). Population density data are from 1km grids from the Center for International Earth Science Information Network (Columbia University) (CIESIN).

Despite their urban concentration, private hospitals operate at a significantly smaller scale than public specialist hospitals and capture limited market share ([Table A.1](#)). The 134 private hospitals in the event study sample average 94 beds, compared to 509 beds for public specialist hospitals and 89 beds for non-specialist public hospitals. The private sector includes large corporate chains such as KPJ

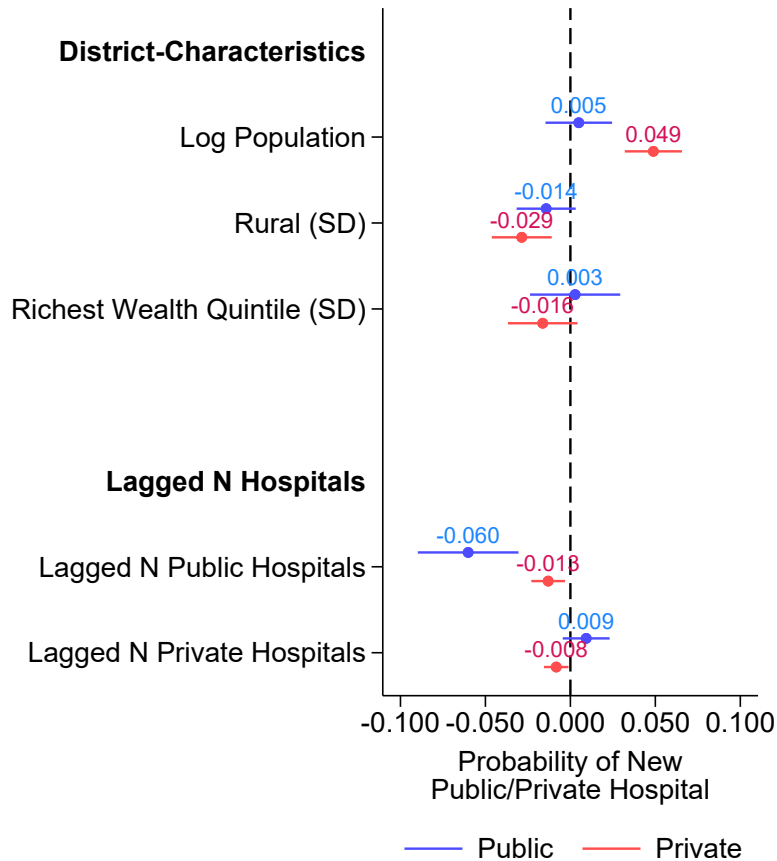
Healthcare (16 percent) and Pantai Holdings (8 percent), but 65 percent of private hospitals are independently owned, typically founded by one or a small group of specialist physicians (Table A.1, Panel A). For most private hospitals in Malaysia, recruiting specialist physicians is a key binding constraint for establishment. The law requires every private hospital to appoint a medically qualified person in charge with specialty training, and small independent hospitals typically depend on their founding physician to fill this role.

In terms of market shares, I use maternity services as an example. Private hospitals hold just 8 percent of district market share for vaginal deliveries, while public specialist and non-specialist hospitals capture 70–79 percent (Table A.1, Panel B). The remaining share reflects the outside option of home births, traditional birth attendants, and private maternity clinics. This small market share occurs despite public specialist hospitals facing congestion: public specialist hospitals have 73.9 percent bed occupancy rates compared to 47.3 percent for public non-specialist hospitals and 53.9 percent for private hospitals (Table A.1, Panel A). The congestion creates wait time dissatisfaction among public hospital patients (3.23 vs 3.82 satisfaction rating for private hospitals), yet survey respondents still rate public hospitals higher on overall quality (4.03 vs 3.83 for private hospitals; Table A.1, Panel C).

The limited overlap between public and private hospitals partly reflects segmented price markets. Private maternity services cost 3,306 MYR compared to 100 MYR for subsidized public services (Table A.1, Panel B). This price gap corresponds to differences in the patient populations they serve. Private hospital users have higher monthly incomes (2.54 vs 1.52 thousand MYR), shorter travel distances to private facilities (15.31 vs 31.82 km), and higher private insurance rates (0.52 vs 0.16; Table A.1, Panel C).

These differences extend to location decisions. [Figure 2](#) presents the average marginal effects of district characteristics on the probability of public and private hospital entry from a logit model with year fixed effects. I omit district fixed effects to compare the characteristics of districts that received a new hospital to those that did not within the same year. The results show that the covariates predicting private entry differ from those predicting public entry. Private hospitals enter districts with higher population and lower proportion of rural residents. Public hospitals, conversely, are significantly less likely to enter districts that already have a public hospital, consistent with the Ministry of Health's stated objective of expanding access to underserved areas rather than duplicating existing public capacity. Factors that strongly predict private entry, such as population and urbanization, show weaker or statistically insignificant associations with public hospital placement. While this descriptive evidence does not rule out all potential confounding, it suggests that public hospital allocations are not primarily driven by the same profit considerations that incentivize private hospital entry.

Figure 2: Descriptive Evidence on Public and Private Hospital Entry



Note: These are average marginal effects from logit regressions of public (or private) hospital entry on a set of district characteristics with year fixed effects. The data consists of public and private entry between 1996 and 2013. The mean probability for public entry is 0.016 while it is 0.024 for private hospitals. The full coefficient plot can be found in [Figure A.5](#). The dependent variable is a binary variable for whether a district-year receives a new public (or private) hospital. Standard errors are clustered at the district level.

4 Conceptual Framework

A new public hospital is both a subsidized competitor and a producer of the private sector's workforce. I organize these two roles in a simple model in the tradition of Bresnahan and Reiss (1991), in which the public hospital is not a strategic player but an exogenous shifter of the profit available to private entrants. The model is static and yields three predictions that guide the empirical analysis and a back-of-the-envelope valuation exercise.

Setup. Consider a geographically distinct healthcare market m of size S_m , measured by population. A pool of identical potential private entrants faces a common sunk cost F of establishing a hospital. The market receives one public hospital, priced below marginal cost by policy and placed exogenously with respect to private entry. The public hospital does not choose price or capacity in response to private firms. It enters the model only through its effect on the profit available to private entrants.

The Public Hospital. I assume the public hospital scales the per-firm variable profit available to a private entrant by a common factor e^δ , and that the log shift δ splits additively into a demand channel and a labor channel,

$$\delta = \underbrace{\delta_D}_{\text{Demand channel}} + \underbrace{\delta_L}_{\text{Labor channel}}, \quad \delta_D \leq 0, \quad \delta_L \leq 0. \quad (1)$$

referrals – diversion
pool depth – hiring

Each channel nets a complementarity against a competition channel. The demand channel δ_D combines referral and agglomeration spillovers, which raise private demand. It also captures competition as patients switch to the subsidized public option. The labor channel δ_L combines a deeper local pool of specialists, who staff

private hospitals at lower cost, against the public hospital's own hiring from that pool. This tension between competition and complementarities is the primary object the paper studies.

Private Entry. Given the public hospital, per-firm variable profit is market size times a per-firm term times the shift,

$$\pi(N) = S_m e^\delta v(N), \quad v'(N) < 0, \quad (2)$$

where N is the number of private entrants and $v(N)$ is per capita variable profit per firm absent a public hospital, so that $\delta = 0$ recovers a market with no public hospital. Population scales profit, v falls in N because each additional entrant divides the residual market and compresses the markup, and e^δ is the public hospital's proportional shift. Firms enter until the marginal entrant cannot cover the sunk cost, so equilibrium entry is

$$N_m = \max\{N : S_m e^\delta v(N) \geq F\}. \quad (3)$$

A market supports at least N private hospitals only if its size exceeds the entry threshold

$$S_N \equiv \frac{F}{e^\delta v(N)}, \quad (4)$$

which rises in N because v falls in N .

Equilibrium Entry. Variable profits are unobserved so I make a parametric assumption. I parameterize per-firm variable profit as constant-elasticity, $v(N) = v_0 N^{-\varepsilon_N}$, where

$$\varepsilon_N \equiv -\frac{d \ln v}{d \ln N} > 0 \quad (5)$$

is the elasticity of per-firm variable profit with respect to the number of entrants. Dropping the integer restriction as a common assumption in this literature (Mankiw and Whinston, 1986), the marginal entrant earns exactly zero, so with the parameterization the zero-profit condition becomes

$$S_m e^\delta v_0 N_m^{-\varepsilon_N} = F, \quad (6)$$

and taking logs and solving for $\ln N_m$ gives the form

$$\ln N_m = \frac{1}{\varepsilon_N} \left(\underbrace{\ln S_m}_{\text{Market size}} + \underbrace{\delta}_{\text{Public hospital}} + \ln v_0 - \ln F \right). \quad (7)$$

Market size and the profit shift enter symmetrically, each divided by ε_N . A proportional shift in per-firm profit therefore has the same effect on entry as a log increase in market size of the same magnitude. Equation (7) gives two responses that the empirical work estimates. Holding the public hospital fixed, the response of log entry to log market size is the entry gradient,

$$\underbrace{\frac{\partial \ln N_m}{\partial \ln S_m}}_{\text{Entry gradient}} = \frac{1}{\varepsilon_N}, \quad (8)$$

so in levels one log point of market size supports N_m/ε_N additional private hospitals. Holding market size fixed, a public hospital that shifts the profit index by δ raises log entry by

$$\Delta \ln N_m \equiv \ln N_m(\delta) - \ln N_m(0) = \frac{\delta}{\varepsilon_N}, \quad (9)$$

the difference between log entry with the public hospital and log entry without the public hospital. Both responses carry the competition elasticity ε_N , which is unobserved. The ratio between $\Delta \ln N_m$ and $\partial \ln N_m / \partial \ln S_m$ cancels it and returns the profit shift,

$$\delta = \frac{\Delta \ln N_m}{\partial \ln N_m / \partial \ln S_m}. \quad (10)$$

Both the numerator and the denominator is estimable from data, so the ratio identifies the public hospital's proportional effect on per-firm variable profit.

Predictions. The framework yields three predictions.

Prediction 4.1 (Crowd-in versus crowd-out). By equation (9), a new public hospital crowds in private entry when its complementary effects outweigh its competitive effects, $\delta > 0$, and crowds out private entry otherwise.

Prediction 4.2 (Market size). The entry response has an extensive and an intensive margin in market size. On the extensive margin, a market below the threshold S_1 in equation (4) holds no private hospital, and a profit shift leaves entry unchanged unless it brings the threshold below market size. On the intensive margin, the level response is $N_m(e^{\delta/\varepsilon_N} - 1)$, which scales with the level of entry and is therefore larger in more populous markets. Both channels concentrate crowd-in in large markets near the private entry margin.

Prediction 4.3 (Valuation). The crowd-in effect and the entry response to market size jointly identify the proportional profit shift a public hospital induces. The sunk cost, market size, and the competition elasticity ε_N enter both responses through the same denominator and cancel in the ratio. Two reduced-form objects, the crowd-in estimate and the entry response to market size, therefore price the public hospital in units of a private entrant's variable profit. [Section 5.4](#) implements this ratio.

Welfare. The model isolates the effect of a public hospital on private entry, and takes no stance on the joint welfare of both sectors. There are two points worth mentioning. First, crowd-in is not itself a welfare gain. Within the private market, free entry with business stealing can be socially excessive, as part of each entrant's variable profit in equation (2) is demand diverted from incumbents rather than surplus created (Spence, 1976; Mankiw and Whinston, 1986). Second, a full welfare evaluation would require the equilibrium response of both sectors. Crowded-in private hospitals serve patients who can pay, while the public hospital guarantees access for all. Evaluating the reallocation means weighing patient welfare across the two options, the distributional value of subsidized public access, and the price and quality adjustments the model holds fixed. These general-equilibrium objects are outside the scope of this paper.

5 Reduced Form Evidence

5.1 Impact of New Public Hospitals on Private Entry

Between 1996 and 2013, Malaysia built new public hospitals across 25 districts in response to public backlash against proposed privatization policies of the public healthcare system. I exploit the staggered timing of public hospital openings across these 25 treated districts and 22 never-treated controls in an event study design ([Figure B.1](#) maps these districts). By 2013, there were a total of 146 and 143 public hospitals across 133 districts. Never-treated districts are those that never received a public hospital over the entire panel. Districts with public hospitals built before 1996 are excluded from the control group as earlier construction may carry dynamic effects into the study period.

A key threat to identification is that the same factors driving private entry may also influence the location of new public hospitals. The Ministry of Health allocates public hospitals based on accessibility, population size, and congestion at existing facilities, criteria that may also drive private entry. However, balance tables between treated and control districts suggest that characteristics are largely similar across the two groups, with some differences in population, age, and rurality ([Table B.1](#)).

I estimate the interaction-weighted event study specification of Sun and Abraham (2021) (SA), which recovers cohort-specific average treatment effects on the treated by comparing each treated cohort to never-treated districts separately. Comparing treated cohorts to later-treated districts yields similar results (see [Table B.9](#)). The SA estimator avoids the contamination bias that arises in conventional two-way fixed effects estimation when treatment effects vary across cohorts (de Chaisemartin and D’Haultfœuille, 2024). Cohort treatment heterogeneity is plausible in this context, as different years may have unique characteristics that affect how public

hospital construction impacts private entry. For the main event study results, I use the SA estimator with robustness checks that include alternative staggered difference-in-differences estimators. Two-way fixed effects results are similar to the SA estimator and are reported in [Table B.7](#).

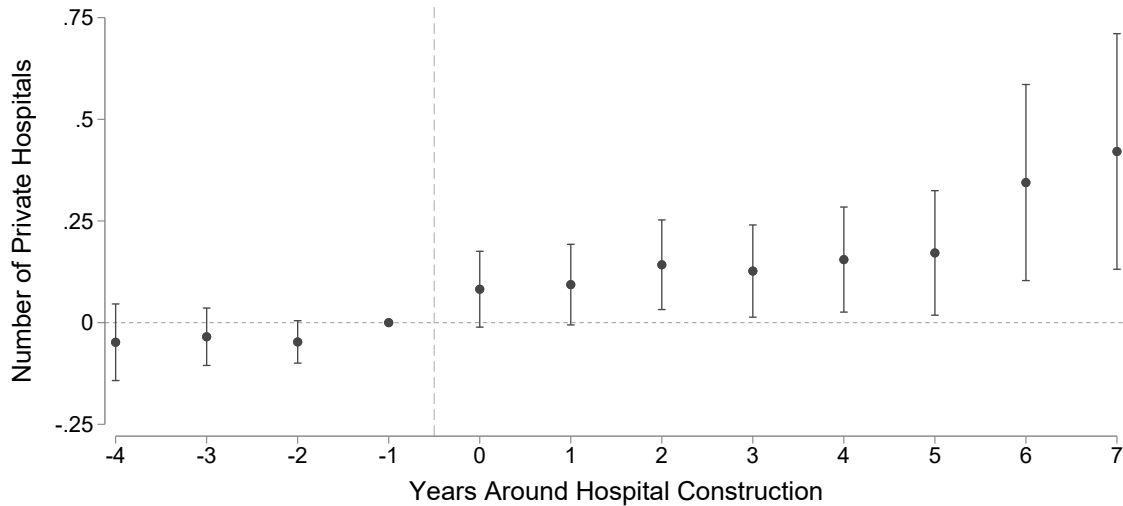
$$Y_{dt} = \mu_d + \lambda_t + \sum_{e \in \mathcal{E}} \sum_{\ell \neq -1} \tau_{e,\ell} \mathbf{1}\{E_d = e\} D_{dt}^\ell + u_{dt} \quad (11)$$

where Y_{dt} is the number of private hospitals in district d at time t , μ_d and λ_t are district and year fixed effects, E_d is the year district d receives its first public hospital, and $\ell = -1$ is the reference period. District fixed effects absorb the time-invariant component of allocation, year fixed effects absorb nationwide healthcare trends, and the effect is identified from within-district changes around the timing of construction (Deshpande and Li, 2019; Cengiz et al., 2019). Three districts received more than one public hospital during 1996–2013. For these, I use the year of the first treatment, and results are robust to excluding them from the sample ([Table B.8](#)).

[Figure 3](#) plots the main event study results. Coefficients are flat and close to zero in the four years preceding construction. The year of construction shows an increase in the number of private hospitals relative to control districts. This effect continues to grow over the subsequent years, reflecting the dynamic response of private hospital entry following public hospital construction. By year seven, treated districts have 0.421 more private hospitals relative to control districts, or 29 percent of the pre-treatment mean. Averaged over the post-treatment window, the effect is 0.465 against a pre-treatment mean of 1.440 ([Table 1](#), Column 1). The dynamic effect suggests that complementarities increase over time, though the effect at $t = 0$ also suggests that some complementarities are realized immediately.

The main effect masks significant heterogeneity by whether a public hospital is a specialist or non-specialist hospital. [Figure 4](#) splits the event study treatment

Figure 3: Effects of New Public Hospitals on Private Hospitals



Note: Event study estimates from [Equation 11](#), truncated to four pre-treatment and seven post-treatment periods. Events consist of 25 newly constructed public hospitals. Each dot represents a point estimate. Vertical lines show 95 percent confidence intervals from clustered standard errors. Full coefficient table in [Table B.2](#).

districts by hospital type. These results plot 14 specialist public hospitals and 11 non-specialist public hospitals separately. Specialist public hospitals crowd in private entry, with coefficients rising from approximately 0.199 at the year of construction to approximately 0.861 by year seven, or 33 percent of the pre-treatment mean. Averaged over the post-treatment window, the specialist effect is 0.785, or approximately 30 percent of the pre-treatment mean ([Table 1](#), Column 2).

I interpret the non-specialist estimates as a null rather than crowd-out. The coefficients hover slightly below zero throughout the post-treatment window, but two features caution against a causal reading. First, these districts hold no private hospitals at baseline ([Table 1](#), Column 3, 'Pre-Treatment Mean'). The negative estimates thus come mechanically from differencing zero-entry treated districts against controls with positive entry. Second, their pre-trends drift down almost linearly before construction.

Table 1: Average Post-Treatment Effects on Private Hospitals

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.465 (0.094)		
Specialist public hospitals		0.785 (0.108)	
Non-specialist public hospitals			−0.171 (0.009)
Mean Dep. Var.	1.303	1.701	0.631
Pre-Treatment Mean	1.440	2.571	0.000
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N Districts $\times$ Year	846	648	594
R^2	0.951	0.954	0.930
Treated Districts	25	14	11
Estimator	SA	SA	SA

Notes: Each column presents results from separate regressions using the Sun and Abraham (2021) interaction-weighted estimator. Coefficients represent the average of post-treatment event study coefficients. The dependent variable is the number of private hospitals in a district. Column (1) pools all public hospitals; columns (2) and (3) separate by type. Mean Dep. Var. is the average outcome across all district-years in the estimation sample, pooling treated and control districts. Pre-Treatment Mean is the average outcome among treated districts in the pre-treatment period. Standard errors clustered at the district level in parentheses. Bootstrap and randomization inference p -values are reported in [Section B.3](#).

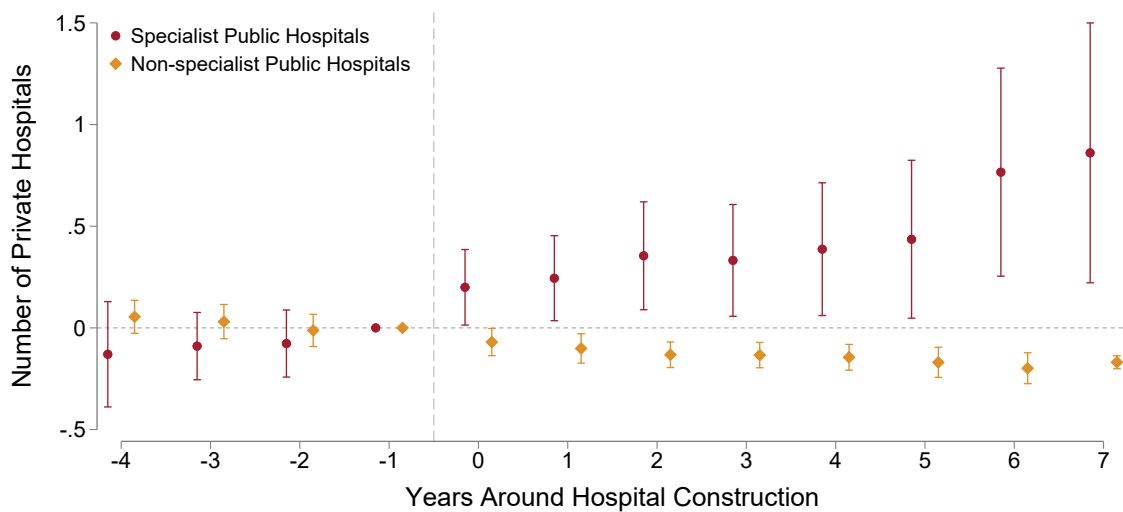
A concern with staggered event studies based on few treatment clusters is that conventional clustered standard errors overstate statistical significance (Conley and Taber, 2011; Ferman and Pinto, 2019). I implement two alternative inference procedures that do not rely on asymptotic approximations and that quantify distinct notions of uncertainty. First, a pairs cluster bootstrap that treats districts as sampled from a larger population and estimates the sampling distribution of the aggregate coefficient. Second, a randomization inference which holds the districts fixed and asks how extreme the estimate is relative to the distribution generated by reassigning treatment across districts under the null of no effect (Abadie et al., 2020).

Details are in [Section B.3](#). In short, the two procedures find that the specialist crowd-in effect is statistically significant, while the non-specialist effect is less precisely estimated.

The result is also robust across alternative specifications. Sensitivity analysis following Rambachan and Roth (2023) shows that the main result survives post-treatment trend deviations close to half of the largest pre-treatment deviation magnitude ([Section B.6](#)). The results hold under synthetic difference-in-differences ([Table B.12](#)), coarsened exact matching ([Figure B.6](#)), exclusion of multiple-treatment districts ([Table B.8](#)), later-treated controls ([Table B.9](#)), and across four alternative staggered DiD estimators ([Table B.10](#)). Weighting by public hospital bed capacity increases the specialist effect, consistent with training capacity scaling with hospital size ([Table B.11](#)). The crowd-in also scales with district size, as [Prediction 4.2](#) predicts: interacting the specialist treatment with pre-treatment population size amplifies the public specialist crowd-in effect ([Table B.18](#)). However, the interactions are identified from 14 specialist-treated events, so I interpret this result with caution.

One key restriction is that the entry variable measures cumulative construction and ignores exit. The hospital panel is built from the commissioning dates of hospitals observed in 2013. A hospital that entered and closed before 2013 never appears, and the stock is non-decreasing by construction. Thus, the effects are a lower bound on private entry. Overall, the main result shows that public hospital crowds-in new private hospitals instead of crowding-out. The growing effect over time on private hospitals hints at the possible mechanisms. While it does not rule out mechanisms that might be one-off, it does support a more persistent, gradual mechanism over time.

Figure 4: Effects of New Specialist and Non-Specialist Public Hospitals on Private Hospitals



Note: Event study estimates from [Equation 11](#), truncated to four pre-treatment and seven post-treatment periods. Events consist of 25 newly constructed public hospitals split into specialist and non-specialist events. Each dot represents a point estimate. Vertical lines show 95 percent confidence intervals from clustered standard errors. Full coefficient table in [Table B.2](#).

5.2 Mechanism: Healthcare Utilization and Physician Supply Spillovers

The heterogeneity above raises a question: through what channel does the specialist crowd-in occur? The demand channel asks whether public hospitals expand or substitute for total healthcare utilization, while the labor channel asks whether public hospitals expand the local stock of trained healthcare professionals. The demand channel uses healthcare utilization data from a nationally representative household survey conducted in 1996, 2006, and 2011⁸. The labor channel uses data from the Malaysian Population Census of 1970, 1980, and 1991, which records occupations at sufficient detail to identify specialist physicians by employment status and sector.⁹ Given that neither is longitudinal with continuous time points, I use a stacked 2x2 difference-in-differences specification rather than the full event study design:

$$Y_{sdt} = \beta \cdot \text{Post}_t \times \text{Treated}_d + \alpha_{ds} + \lambda_{ts} + u_{sdt} \quad (12)$$

where α_{ds} are district-by-stack fixed effects and λ_{ts} are year-by-stack fixed effects. Treated districts in the utilization stacks are those first receiving a public hospital of the relevant type during each stack's window (1997–2006 and 2007–2011). Districts treated during the first window are excluded from the second stack, and the control

⁸Utilization rates are self-reported among residents living within a district and might include cross-district utilization. This explains why the pre-treatment utilization mean for non-specialist hospitals is non-zero despite having zero pre-treatment private hospitals.

⁹The 2000 census adopted revised ISCO occupation codes that combine physicians, veterinarians, dentists, and other medical professionals into a single category, precluding identification of specialist physicians. I therefore cannot measure district-level physician supply during the study period itself (1996–2013) and rely on pre-period census evidence supplemented by contemporary registry data. I identify private specialist physicians by 'self-employment' category as most private specialist physicians in Malaysia are fee-for-service independent practitioners, not salaried hospital staff. This overcounts the number of specialist physicians as it also includes physicians that practice in the private sector.

group in each stack comprises never-treated districts together with districts not yet treated by the stack's end year.

Healthcare Utilization: Treatment Substitution or Expansion? [Table 2](#) tests whether public hospitals expand the total healthcare market or substitute for private care. Panel A shows that specialist hospitals do not affect total inpatient admissions or total outpatient visits (Columns 1 and 3), while reducing private inpatient admissions by nearly half relative to the pre-treatment baseline (Column 2) and private outpatient visits by more than a third (Column 4). This result, in which public utilization grows at private utilization's expense without expanding total utilization, is consistent with patient substitution toward the new public option. Panel B shows that non-specialist hospitals produce directionally similar but imprecise reductions across all utilization measures. Two additional facts reinforce the notion that demand spillovers do not explain the crowd-in effects. First, cross-sector referrals are negligible, as private-to-public referrals account for only 0.05 percent of public hospital admissions in 2013 public electronic health records. In addition, specialist hospitals do not increase total outpatient volume (Column 3), ruling out a congestion channel that would occur through overflow from saturated primary care.

Table 2: Effects of Public Hospitals on Healthcare Utilization

	Inpatient Admissions (in 10,000)		Outpatient Visits (in 10,000)	
	Total (1)	Private (2)	Total (3)	Private (4)
<i>Panel A. Specialist Public Hospitals</i>				
Post $\times$ Treated	−0.167 (0.330)	−0.299 (0.140)	0.086 (0.659)	−1.371 (0.671)
Mean Dep. Var.	1.409	0.428	2.008	1.630
Pre-Treatment Mean	2.610	0.651	4.333	3.703
Observations	120	120	120	120
R^2	0.871	0.815	0.865	0.933
<i>Panel B. Non-Specialist Public Hospitals</i>				
Post $\times$ Treated	−0.369 (0.228)	−0.111 (0.099)	−0.515 (1.027)	−0.304 (0.231)
Mean Dep. Var.	0.842	0.293	1.511	0.880
Pre-Treatment Mean	0.465	0.022	2.705	0.490
Observations	114	114	114	114
R^2	0.834	0.804	0.750	0.942

Notes: Stacked difference-in-differences estimates using equation (12). Each column reports the average treatment effect of public hospital construction on healthcare utilization from the National Health and Morbidity Survey (1996, 2006, 2011). Panel A: specialist public hospitals (14 treated districts). Panel B: non-specialist public hospitals (11 treated districts). Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Standard errors clustered by district in parentheses.

Private Specialist Physician Supply. The utilization results presents a puzzle: specialist hospitals reduce private admissions by nearly half and private outpatient visits by more than a third, yet private hospitals still enter. If competition on the demand side is strong, the supply side must offer a counteracting force. In principle, a public hospital deepens the local supply of every profession it trains, from nurses to allied health workers to physicians. Two features single out specialist physicians as the input that governs private entry. First, specialists take far longer to train than nurses or technicians, so the local specialist stock adjusts slowly and a new training hospital relaxes this constraint. Second, a private hospital cannot open without specialist physicians: Malaysian licensing conditions approval on specialist staffing appropriate to the services the hospital offers.¹⁰

Specialist hospitals more than double the total physician stock in treated districts (Column 1) and nearly triple the stock of private specialist physicians (Column 3), raising the private specialist share of total physicians by approximately 28 percentage points from a pre-treatment mean of 38 percent (Column 5). The lag structure traces the training pipeline directly: at lag 0, the immediate effect on the private specialist stock is approximately 55 physicians, growing to approximately 108 by lag 3 as training cohorts graduate and transition to private practice (Table B.4). Non-specialist hospitals, which are staffed primarily by nurses and general practitioners, show no effect on any physician outcome (Columns 2, 4, 6).

Private entrants in the 2013 hospital census carry a median of 20 doctors and a mean of 30. Staffing the specialist crowd-in of 0.785 hospitals therefore requires 16 to 23 specialists per treated district. The estimated inflow is 108, against a pre-treatment stock of 57. The gap reflects the breadth of the self-employment measure.

¹⁰Under the Private Healthcare Facilities and Services Act 1998 (Act 586) and its 2006 regulations, approval to operate a private hospital is conditioned on proposed specialist manpower commensurate with the services offered, so local specialist availability is a prerequisite for licensed entry.

Self-employment captures physicians who move to the private side in any form of practice, including specialists who open clinics and office-based practices, not only hospital consultants.

In addition, the spillover is specific to specialists. Specialist hospitals raise nurse and technician headcount at the new facility, but neither profession shows a similar result into self-employed private practice ([Table B.17](#)).

Table 3: Effects of Public Hospitals on Specialist Physician Supply (3 Years Post-Construction)

	Total Physicians (in 1,000s)		Self-Employed Specialists (in 1,000s)		Self-Employed Share of Total	
	(1)	(2)	(3)	(4)	(5)	(6)
Specialist	0.192 (0.117)		0.108 (0.047)		0.275 (0.135)	
Non-specialist		-0.024 (0.027)		-0.008 (0.006)		-0.022 (0.015)
Mean Dep. Var.	0.115	0.054	0.041	0.014	0.218	0.093
Pre-Treatment Mean	0.150	0.000	0.057	0.000	0.375	0.000
District $\times$ Stack FE	Yes	Yes	Yes	Yes	Yes	Yes
Year $\times$ Stack FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	62	66	62	66	62	66
R^2	0.863	0.923	0.858	0.880	0.940	0.985

Notes: Stacked difference-in-differences estimates using census data (1970, 1980, 1991). Effects measured 3 years post-construction to allow specialist training programs to mature; lag robustness is in [Table B.4](#). Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Non-specialist treated districts have zero pre-treatment physician counts as these hospitals are constructed in areas without existing medical infrastructure. Standard errors clustered by district in parentheses.

Descriptive Evidence from Individual Physician Registry Data. The census evidence measures the stock of physicians but cannot directly observe the flow of specialists from public training to private practice. I provide direct individual-level evidence on this flow using Malaysia’s Annual Practicing Certificate (APC) registry, a mandatory annual license covering 12,503 registered specialists observed from by 2025.¹¹

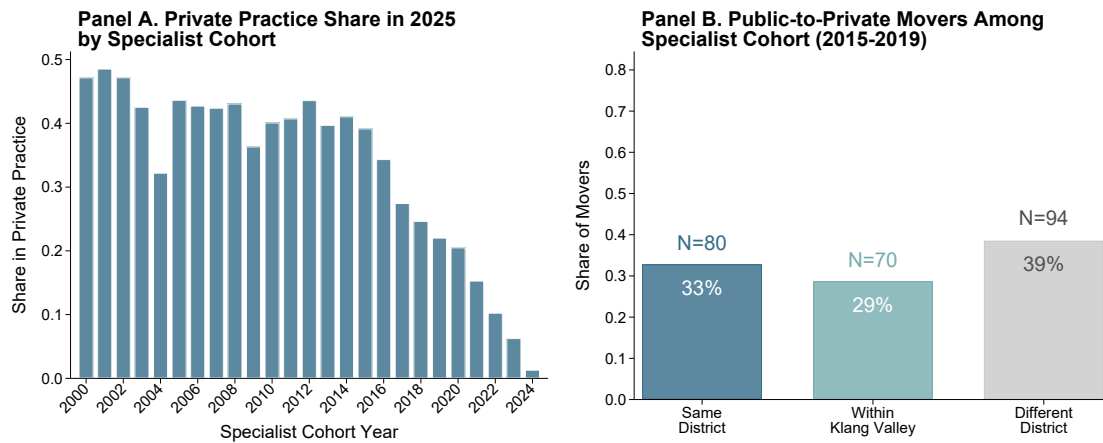
Figure 5 presents two facts supporting the mechanism. Panel A shows the share of specialists in private practice by qualification cohort. Among recent cohorts (2022–2024), fewer than 10 percent practice privately, consistent with the 4–7 year compulsory service bond following government-sponsored training. As bonds expire for earlier cohorts, the private practice share rises steeply, reaching approximately 40 percent for cohorts qualifying before 2015. This implies a long-run steady state in which approximately two in five specialists eventually transition to private practice.

Panel B shows the geographic concentration of public-to-private transitions among 244 specialists from 2015–2019 cohorts observed making this move within 2023 through 2026. One-third remain in the same district as their last public posting, and an additional 29 percent remain within the greater ‘Klang Valley’ metropolitan area which encompasses the capital city of ‘Kuala Lumpur’ and multiple smaller districts.¹² Together, approximately 62 percent of movers remain in the same local labor market, consistent with specialist hospitals expanding the local supply of trained specialists who subsequently establish or join nearby private practices. The census evidence in Table 3 covers 1970–1991, predating the main study period

¹¹Specialist qualification year refers to the year in which the specialist degree (e.g., Master of Medicine) is conferred by the training institution. The APC data records each physician’s primary practice location, sector (public, private, university, military), and any secondary practice locations for each year of active licensure.

¹²The Klang Valley encompasses Kuala Lumpur, Petaling, Hulu Langat, Gombak, Klang, Putrajaya, and Kuala Langat.

Figure 5: Direct Evidence on the Public-to-Private Specialist Physician Pipeline



Note: Panel A shows the share of specialists currently in private practice by year of specialist qualification, using 12,503 registered specialists from Malaysia’s Annual Practicing Certificate (APC) registry (2025). Panel B classifies 244 public-to-private transitions among specialists from 2015–2019 qualification cohorts by geographic proximity to their last public hospital posting.

of 1996–2013. Two institutional facts suggest that the training channel is valid throughout the study period. First, the legal framework has been the same since 1971. The Medical Act 1971 establishes compulsory public service and annual practicing certificate requirements that have remained in place, and the government scholarship that funds specialist training with a compulsory service bond has operated since then¹³. Second, the APC registry shows the pipeline remains active in 2025, with approximately 40 percent of specialists eventually transitioning to private practice.

Hospital Upgrade Falsification Test. Between 2003 and 2012, 35 public hospitals underwent significant bed capacity expansions while maintaining their status as specialist or non-specialist hospitals. In the framework’s terms, an upgrade

¹³The first significant disruption being the introduction of contract employment in 2017, after the study period (Ismail et al., 2025). The only major regulatory amendment, in 2012, introduced a formal National Specialist Register. Before this, no specialist registration system existed, and transitions from public to private practice required only an Annual Practicing Certificate. The pipeline was therefore less regulated during the study period than it is today.

shifts the demand channel while leaving the labor channel unchanged. [Table B.14](#) shows that specialist hospital upgrades have limited effects on private entry, as the framework predicts when the labor channel is absent.

5.3 Within-District Spatial Evidence: Competition versus Complementarity

The mechanism above shows both competition and complementarity channels occur at the district level. Here, I use a Hotelling model intuition to show the tension between competition and complementarities in how private entrants locate. Specifically, I exploit within-district variation in the distance between new specialist public hospitals and private entrants.

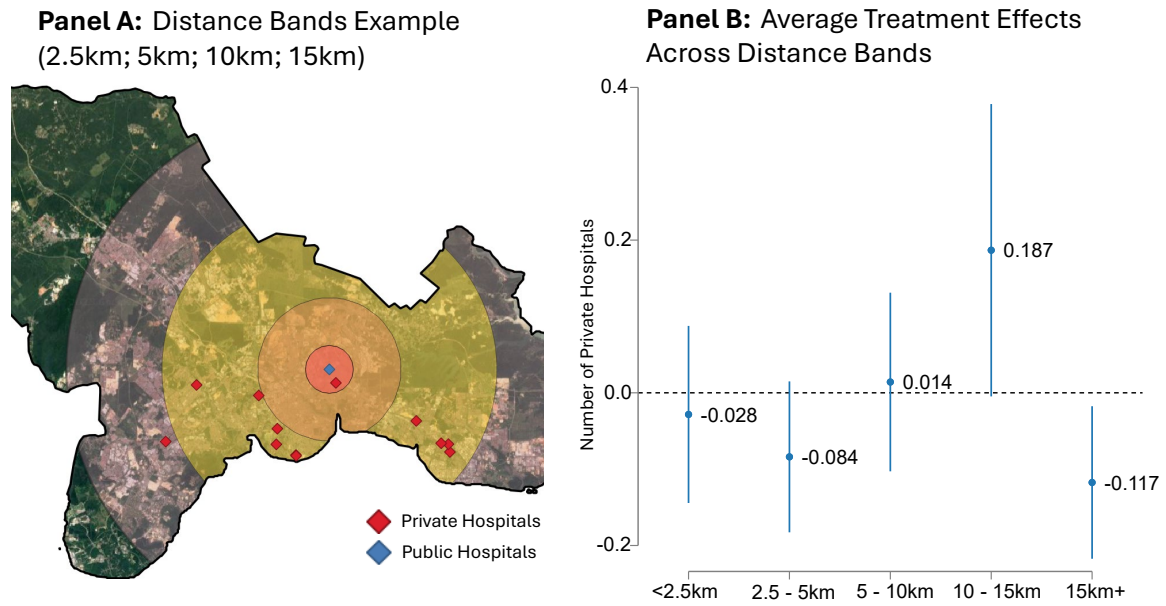
In theory, competition would incentivize private entrants to locate far from the new public hospital. Physician supply complementarity would show a different pattern as physicians may reside all over the district. Thus, one would expect the private entry response to be non-monotonic in distance: entry should be lowest in the immediate vicinity where competition is intense, highest at an intermediate distance where the labor pool remains accessible, and falling again at greater distances where commuting costs limit physician access.

To capture this spatial gradient, I estimate the same event study specification as in [Equation 11](#) separately for each of five mutually exclusive distance bands b from the new specialist public hospital: 0–2.5km, 2.5–5km, 5–10km, 10–15km, and beyond 15km,

$$Y_{dt}^b = \mu_d^b + \lambda_t^b + \sum_{e \in \mathcal{E}} \sum_{\ell \neq -1} \tau_{e,\ell}^b \mathbf{1}\{E_d = e\} D_{dt}^\ell + u_{dt}^b, \quad (13)$$

where Y_{dt}^b is the number of private hospitals within band b of the new hospital in treated districts, and the number of private hospitals in the entire district in never-treated districts. All other terms are as in [Equation 11](#). Each band regression uses the same control group as the main specification, so the coefficients are comparable across bands. The specification is estimated on the 14 specialist-treated districts only.

Figure 6: Effects of Specialist Public Hospital on Private Entry by Distance



Notes: Panel A maps example distance bands for the Johor Bahru district. The rings represent 2.5km, 5km, 10km and 15km from the newly constructed public hospital. Districts with multiple specialist public hospitals are omitted. Panel B plots post-treatment estimates from [Equation 13](#) across five distance bands. Coefficients are tabulated in [Table B.19](#). The dependent variable in each regression is the number of private hospitals within the specified distance band in treated districts; the control group counts all private hospitals across the entire district in never-treated districts. Pre-treatment mean number of private hospitals: 0.29, 0.29, 1.43, 0.21, and 0.36 for bands 1 through 5. Standard errors clustered at the district level.

The spatial pattern in [Figure 6](#) Panel B traces the predicted non-monotonic gradient across the five distance bands. Private entry is negative in the immediate

0-5km vicinity, with point estimates of approximately -0.028 in the 0-2.5km band and approximately -0.084 in the 2.5-5km band, consistent with patient competition being most intense at short distances ([Table B.19](#)).

The 5-10km band—which contains the highest pre-treatment private hospital density—shows essentially zero effect. Private entry then peaks in the 10-15km band, where private hospitals can presumably access the shared physician labor pool while avoiding direct patient competition. Beyond 15km, the effect turns negative again at approximately -0.117, consistent with commuting costs limiting physician access at greater distances. The result is robust to an alternate design where rings are drawn beyond district lines (see [Table B.20](#)).

5.4 The Value of Public Hospitals for Private Entrants

Here, I extend the reduced form results to address two questions. First, what is a specialist public hospital worth to a private entrant, and second, how much more crowd-in would an alternate placement of the same hospitals have yielded. The conceptual framework provides a simple way to calculate the value of a public hospital by assuming variable profit scales with population. Private entry's response to public hospitals measured against population would then identify the proportional effects on profits (equation (10)). The exercise also inherits the framework's other assumptions: entrants are identical and face a common sunk cost, entry is a one-time decision, and the model is static.

The Entry-Population Gradient. I first estimate the number of additional private hospitals that one log point of population predicts. I fit an entry function on the

census panel of 133 districts observed in 1970, 1980, and 1991,

$$\mathbb{E}[N_{dt} \mid S_{dt}] = \exp(\alpha_t + \beta \ln S_{dt}), \quad (14)$$

a Poisson regression of the private hospital count N_{dt} on log population with census-year effects α_t . I use the census panel rather than the event-study sample, as it covers a wider set of districts and provides a more precise gradient. The coefficient on log population is 2.433, the empirical counterpart of the entry gradient $1/\varepsilon_N$ in equation (8), so a one percent rise in population raises predicted entry by 2.4 percent (Table B.21). Re-estimating on the 47 event-study districts gives 1.890 (Column (3)), which is similar in magnitude.

What a Public Hospital Is Worth. To estimate how much a specialist public hospital is worth to private entrants, I calculate the two components in equation (10). The denominator is the elasticity of entry with respect to market size, which under equation (14) is constant and equal to the coefficient on log population. Taking logs, $\ln \mathbb{E}[N_{dt}] = \alpha_t + \hat{\beta} \ln S_{dt}$, so $\partial \ln N / \partial \ln S = \hat{\beta} = 2.433$. The numerator is the log change in entry the public hospital induces: the crowd-in estimate moves mean predicted entry among the 14 treated districts from $\bar{N} = 2.237$ to $\bar{N} + 0.785$, so $\Delta \ln N = \ln(1 + 0.785/2.237) = 0.301$. Equation (10) then gives $\hat{\delta} = 0.301/2.433 = 0.124$, with a 95 percent confidence interval of 0.095 to 0.152. All in all, a specialist public hospital is worth about 12 percent of a marginal entrant's variable profit, net of the demand loss.

This estimate combines two opposing channels. The demand loss is 37 to 46 percent of baseline private volume (Table 2), and treating a proportional volume loss as a proportional profit loss, equation (1) recovers the labor channel as the net value plus the demand loss. The implied labor-cost saving is 49 to 58 percent of

baseline variable profit at the headline estimate. Assuming a sunk cost of 55 beds at RM 600,000 to 700,000 per bed, a private hospital costs RM 33 to 38.5 million (8.1 to 9.4 million USD) to build.¹⁴ The labor-cost saving is then RM 16 to 22 million per entrant.

Public Hospital Reallocation. Next, I ask how much more crowd-in the same set of hospitals would have yielded under an alternate placement. I predict district-level crowd-in from the entry function. The profit shift $\hat{\delta}$ raises district d 's predicted entry from the fitted count $\hat{N}_d$ to $\hat{N}_d e^{\hat{\beta}\hat{\delta}}$, so the predicted crowd-in is $\hat{\tau}_d = \hat{N}_d (e^{\hat{\beta}\hat{\delta}} - 1)$. The observed allocation yielded a predicted crowd-in of 10.99 hospitals. Six of the 14 districts yielded close to zero predicted private hospitals as those districts have smaller population size. Holding the number of hospitals fixed, I reassign these six to the six highest-predicted candidate districts. The reassignment adds 3.64 hospitals at the new sites against 0.13 forgone at the vacated sites, a net gain of 3.51 hospitals (Table B.16), roughly a third above the status quo. In capacity terms it adds 193 private beds, equivalent to RM 58 million (14 million USD) of public hospital construction at the Malaysia Plan mid-period cost of RM 300,000 per bed.

The gain concentrates in one district. 'Petaling' alone accounts for 2.84 of the 3.64 hospitals added, and excluding it from the candidate pool cuts the net gain to 0.73 hospitals. This concentration is the entry function's own prediction, since predicted entry is convex in population and 'Petaling' is the largest candidate market. It is worth noting that crowded-in private beds are not public beds. Private hospitals charge prices that exclude lower-income patients, and the service mix differs across sectors. The reallocation raises total capacity by moving public capacity away from

¹⁴Documented private projects near the end of the study period: a 100-bed KPJ hospital in Sarawak at RM 70 million, and two private hospitals in Miri at RM 600,000 to 700,000 per bed. The 2010 government benchmark for hospital construction was RM 500,000 per bed.

the districts that held no private hospital at baseline, which are the districts where the public option is the only option.

6 Conclusion

This paper shows that public provision can crowd-in private investment when public providers produce inputs that complement private firms. In Malaysia’s hospital market, specialist public hospitals train specialist physicians that crowd-in private entrants, while non-specialist public hospitals that lack these programs do not. Using hospital surveys, administrative data, census, physician registry, and within-district spatial evidence, I show that the crowd-in occurs in larger districts where private demand can support new entry, and occurs through expanded local specialist physician supply.

The literature on public-private market interactions has documented a variety of equilibrium responses, but most of this work has the public sector acting on the output market, as a substitute, competitor, or quality benchmark (Cutler and Gruber, 1996; Atal et al., 2024; Andrabi et al., 2024; Wilson, 2025). The mechanism in this paper acts on the input side. The public sector holds a monopoly on the production of a complementary input that private firms require for entry. The closest precedent is Bergeaud et al. (2025), in which researcher mobility from public labs generates a knowledge spillover to private R&D, though the spillover there flows through ideas embedded in workers rather than through a public sector labor monopoly.

This labor complementarity is not unique to hospitals. Public legal aid offices train litigators who move into private practice; public school systems develop the teachers whom private schools hire while competing for their students. In each case,

public provision competes with private firms for demand while providing labor supply, and the net effect on private investment depends on which force dominates. The condition that gives the training channel its magnitude is a public monopoly or near-monopoly on training, which leaves private firms unable to substitute away from the publicly trained workforce.

These findings reveal an equity-efficiency tradeoff for allocating scarce public hospital construction budgets across many countries with public-private hospital markets. A type-specific allocation rule that targets specialist hospitals toward districts with high private entry probability and non-specialist toward underserved areas would have crowded-in more private hospital capacity than the observed allocation, at no additional public construction cost. The crowded-in capacity is not free of distributional consequences, however. Private hospitals enter only where willingness-to-pay supports profitable entry, which excludes populations who cannot afford private prices. This implies that the allocation that maximizes total beds concentrates the induced capacity in markets that are already partially served. If the planner's objective is purely to expand access in underserved areas, the observed equity-driven allocation is preferable. If total capacity matters, the type-specific approach yields more private entrants.

This paper takes a positive, not normative stance. The main result identifies a channel through which public provision generates private spillovers, without claiming the resulting crowd-in is welfare enhancing. Two gaps stand between this and a full welfare assessment. First, I do not observe healthcare quality, so if physician migration to private practice degrades public hospitals, the gains from private expansion may come at the expense of populations who rely on public facilities. Second, I document the labor-supply spillover without modeling the physician labor market itself. Understanding how bond length, training capacity,

and geographic restrictions jointly determine the spillover's magnitude would clarify when public provision stimulates complementary private investment and when the physician outflow undermines the public system.

References

- Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2020). Sampling-Based versus Design-Based Uncertainty in Regression Analysis. *Econometrica*, 88(1):265–296.
- Alexander, D. and Richards, M. R. (2023). Economic consequences of hospital closures. *J. Public Econ.*, 221:104821.
- Andrabi, T., Bau, N., Das, J., Karachiwalla, N., and Ijaz Khwaja, A. (2024). Crowding in Private Quality: The Equilibrium Effects of Public Spending in Education*. *Q. J. Econ.*, 139(4):2525–2577.
- Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic Difference-in-Differences. *Am. Econ. Rev.*, 111(12):4088–4118.
- Atal, J. P., Cuesta, J. I., González, F., and Otero, C. (2024). The Economics of the Public Option: Evidence from Local Pharmaceutical Markets. *Am. Econ. Rev.*, 114(3):615–644.
- Banerjee, A., Chowdhury, A., Das, J., Hammer, J., Hussam, R., and Mohpal, A. (2024). The Market for Healthcare in Low Income Countries. *Work. Pap.*
- Barraclough, S. (1997). The growth of corporate private hospitals in Malaysia: Policy contradictions in health system pluralism. *Int. J. Health Serv. Plan. Adm. Eval.*, 27(4):643–659.
- Barraclough, S. (2000). The Politics of Privatization in the Malaysian Health Care System. *Contemp. Southeast Asia*, 22(2):340–359.

- Bergeaud, A., Guillouzouic, A., Henry, E., and Malgouyres, C. (2025). From Public Labs to Private Firms: Magnitude And Channels of Local R&D Spillovers*. *Q. J. Econ.*, 140(4):3233–3282.
- Bloom, N., Propper, C., Seiler, S., and Van Reenen, J. (2015). The impact of competition on management quality: Evidence from public hospitals. *Rev. Econ. Stud.*, 82(2):457–489.
- Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting Event-Study Designs: Robust and Efficient Estimation. *Rev. Econ. Stud.*, page rdae007.
- Bresnahan, T. F. and Reiss, P. C. (1991). Entry and Competition in Concentrated Markets. *J. Polit. Econ.*, 99(5):977–1009.
- Callaway, B. and Sant’Anna, P. H. (2021). Difference-in-Differences with multiple time periods. *J. Econom.*, 225(2):200–230.
- Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Rev. Econ. Stat.*, 90(3):414–427.
- Casadesus-Masanell, R., Nalebuff, B. J., and Yoffie, D. (2007). Competing Complements.
- Cengiz, D., Dube, A., Lindner, A., and Zipperer, B. (2019). The Effect of Minimum Wages on Low-Wage Jobs*. *Q. J. Econ.*, 134(3):1405–1454.
- Conley, T. G. and Taber, C. R. (2011). Inference with “Difference in Differences” with a Small Number of Policy Changes. *Rev. Econ. Stat.*, 93(1):113–125.
- Cremer, H., Marchand, M., and Thisse, J.-F. (1991). Mixed oligopoly with differentiated products. *Int. J. Ind. Organ.*, 9(1):43–53.

- Cutler, D. M. and Gruber, J. (1996). Does public insurance crowd out private insurance? *Q. J. Econ.*, 111(2):391–430.
- Das, J., Hammer, J., and Leonard, K. (2008). The Quality of Medical Advice in Low-Income Countries. *J. Econ. Perspect.*, 22(2):93–114.
- de Chaisemartin, C. and D’Haultfœuille, X. (2024). Difference-in-Differences Estimators of Intertemporal Treatment Effects. *Rev. Econ. Stat.*, pages 1–45.
- De Fraja, G. and Valbonesi, P. (2009). Mixed Oligopoly: Old and New. Discussion Papers in Economics 09/20, Division of Economics, School of Business, University of Leicester.
- Deshpande, M. and Li, Y. (2019). Who is screened out? Application costs and the targeting of disability programs. *Am. Econ. J. Econ. Policy*, 11(4):213–248.
- Dinerstein, M. and Smith, T. D. (2021). Quantifying the Supply Response of Private Schools to Public Policies. *Am. Econ. Rev.*, 111(10):3376–3417.
- Duggan, M. and Scott Morton, F. M. (2006). The Distortionary Effects of Government Procurement: Evidence from Medicaid Prescription Drug Purchasing*. *Q. J. Econ.*, 121(1):1–30.
- Epple, D. and Romano, R. E. (1998). Competition between Private and Public Schools, Vouchers, and Peer-Group Effects. *Am. Econ. Rev.*, 88(1):33–62.
- Ferman, B. and Pinto, C. (2019). Inference in Differences-in-Differences with Few Treated Groups and Heteroskedasticity. *Rev. Econ. Stat.*, 101(3):452–467.
- Gaynor, M., Ho, K., and Town, R. (2014). The Industrial Organization of Health Care Markets. Working Paper 19800, National Bureau of Economic Research.

- Gaynor, M., Laudicella, M., and Propper, C. (2012). Can governments do it better? Merger mania and hospital outcomes in the English NHS. *J. Health Econ.*, 31(3):528–543.
- Ghislandi, S., Renner, A.-T., and Varghese, N. E. (2025). The impact of budget cuts on individual patient health: Causal evidence from hospital closures. *J. Health Econ.*, 101:102975.
- Gruber, J. and Simon, K. (2008). Crowd-out 10 years later: Have recent public insurance expansions crowded out private health insurance? *J. Health Econ.*, 27(2):201–217.
- Herr, A. (2011). Quality and Welfare in a Mixed Duopoly with Regulated Prices: The Case of a Public and a Private Hospital. *Ger. Econ. Rev.*, 12(4):422–437.
- Ho, K. and Lee, R. S. (2017). Insurer competition in health care markets. *Econometrica*, 85(2):379–417.
- Hoxby, C. M. (2000). Does Competition among Public Schools Benefit Students and Taxpayers? *Am. Econ. Rev.*, 90(5):1209–1238.
- Ismail, H., Yacob, M. A., Abdullah, M. H., Zun, A. B., Osman, S., Amran, S. B., Idris, N. A., Fakhru Ruzi, F. R., Jalaluddin, S. N., Abd Kadir, B., Che Arshad, F., and Kamarulzaman, N. A. H. (2025). The Impact of the Amendments of the Medical Act 1971 [Act 50] in 2024 on the Implementation of Specialist Training Programmes by the Ministry of Health. *Malays. J. Med. Sci. MJMS*, 32(3):1–13.
- Jiménez Hernández, D. and Seira, E. (2022). Should the Government Sell You Goods? Evidence from the Milk Market in Mexico.
- Juhász, R., Lane, N., and Rodrik, D. (2024). The New Economics of Industrial Policy. *Annu. Rev. Econ.*, 16(Volume 16, 2024):213–242.

- Kline, P. and Moretti, E. (2014). People, Places, and Public Policy: Some Simple Welfare Economics of Local Economic Development Programs. *Annu. Rev. Econ.*, 6(Volume 6, 2014):629–662.
- Klumpp, T. and Su, X. (2019). Price–quality competition in a mixed duopoly. *J. Public Econ. Theory*, 21(3):400–432.
- Mankiw, N. G. and Whinston, M. D. (1986). Free Entry and Social Inefficiency. *RAND J. Econ.*, 17(1):48–58.
- Rambachan, A. and Roth, J. (2023). A More Credible Approach to Parallel Trends. *Rev. Econ. Stud.*, 90(5):2555–2591.
- Saltzman, E. (2023). What Does a Public Option Do? Evidence from California.
- Shepard, M. (2022). Hospital Network Competition and Adverse Selection: Evidence from the Massachusetts Health Insurance Exchange. *Am. Econ. Rev.*, 112(2):578–615.
- Spence, M. (1976). Product Differentiation and Welfare. *Am. Econ. Rev.*, 66(2):407–414.
- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J. Econom.*, 225(2):175–199.
- WHO (2020). *Private Sector Landscape in Mixed Health Systems*. World Health Organization, Geneva, 1st ed edition.
- Wilson, K. (2025). Does Public Competition Crowd Out Private Investment? Evidence from Municipal Provision of Internet Access. *Am. Econ. J. Microecon.*, 17(4):399–431.

Young, A. (2019). Channeling fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *Q. J. Econ.*, 134(2):557–598.

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A Details on Data and Context

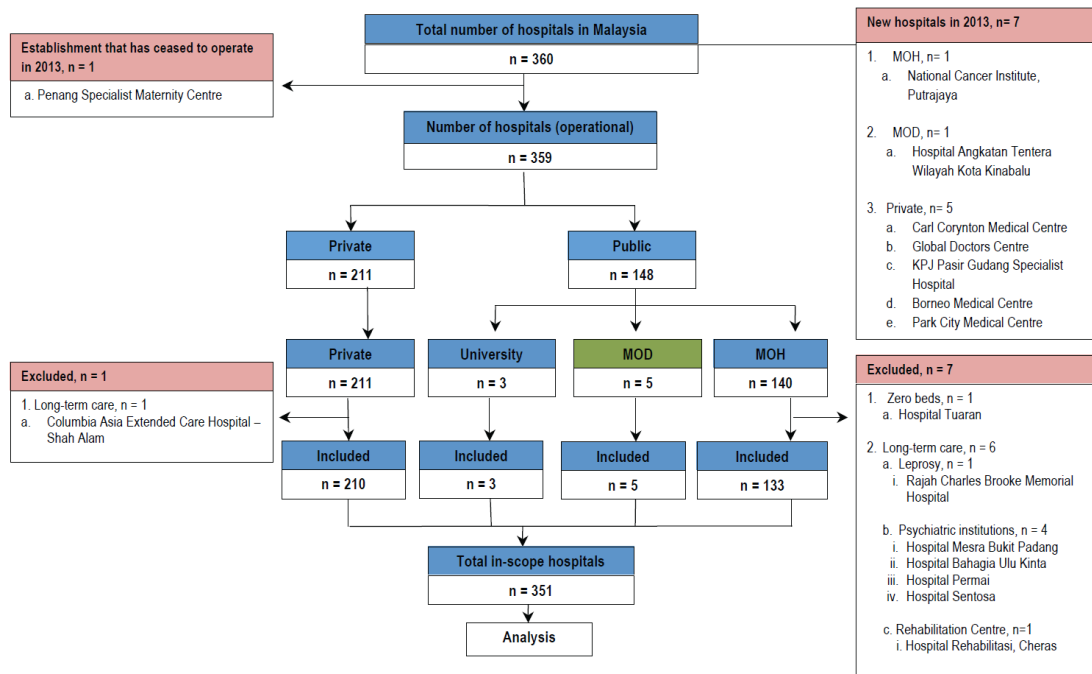
A.1 Hospital Survey Data Details

Hospital survey data comes from the National Healthcare Establishment and Workforce Survey (NHEWS) (2010-2013) conducted by the Clinical Research Centre. This survey provides me with a panel dataset of hospitals, whether the hospital provides certain services, the year in which hospitals began providing services, and year-specific levels of admission, congestion.

The survey gathers information on hospitals in the country regarding their services, with emphasis on specialized clinical services, facilities, medical devices, and health workforce. The NHEWS survey covers all acute curative hospitals and related specialty services for both public and private sectors. This survey asks all facilities that provide inpatient admissions in Malaysia. The survey respondent is the person-in-charge for the administrative department of hospitals. Response rates for public hospitals are 100 percent but for private hospitals it is 83.6 percent for those with less than 20 medical subspecialties and close to 90 percent for those with more than 20 medical subspecialties.

Figure A.1: Sample Details for NHEWS (2010-2013)

CONSORT DIAGRAM NHEWS 2013 (ACUTE CURATIVE HOSPITALS)



Notes: This figure presents sample details from the National Healthcare Establishment and Workforce Survey (NHEWS) in 2013.

A.2 Additional Survey Data Details

The National Health and Morbidity Survey is a nationally representative, two-stage (states and urban-rural status) stratified randomly sampled household survey in Malaysia. The survey asks respondents on hypothetical choice scenarios for birth deliveries after sociodemographic questions—which allows me to observe quality ratings and waiting time ratings. Specifically, the question asks:

“Which is the main health facility you would go to in the following situations?” “For birth delivery, where would you go?”.

The respondent could choose exactly one from four possible responses: “government”, “private”, “traditional/complementary/alternative health facility” and “will not go to any facility”. Following the hypothetical choice questions, the survey asks individuals on their perception of quality. The survey question asks the respondent on ratings on a 1 to 5 Likert scale, which encompasses 12 different questions of quality for outpatient care and inpatient care, in both the public and the private sector. I specifically use the questions *‘Based on your perception or impression, how would you rate the government and private hospital on the following aspect ...’*. First, *‘The waiting time to see a doctor once arrived at a hospital’* followed by *‘Your overall impression’*. Answers do not correspond to specific types of health conditions, and instead refers generally to the public and private sector.

Figure A.2: Quality Perception Survey Questionnaire

 Sangat tidak bagus Very Poor 1		 Tidak bagus Poor 2		 Sederhana Fair 3		 Bagus Good 4		 Sangat bagus Excellent 5	
		(-7) TT		(-9) EJ					
Berdasarkan tanggapan atau kepercayaan anda [kepada penemuramah: sekiranya responden menghadapi masalah, anda boleh bantu dengan mencadangkan, e.g. Daripada perkhabaran/pengalaman keluarga, rakan-rakan anda, pengalaman anda sendiri], bagaimana anda menilai HOSPITAL kerajaan dan swasta (pesakit dalam) pada aspek berikut ... / <i>Based on your perception or impression [to interviewer: if respondent has trouble answering, you can help them by suggesting, e.g. What you hear from your relatives and friends, other's experience, own experience], how would you rate the government and private HOSPITAL (inpatient) on the following aspect ...</i>									
Tanya semua soalan berkenaan fasiliti Kerajaan dahulu, diikuti dengan Swasta.									
					Hospital / Hospital				
					Kerajaan Government		Swasta Private		
AC215	Kesesuaian lokasi hospital <i>Convenience of hospital location</i>								
AC216	Boleh memohon bilik persendirian / tidak berkongsi dengan ramai pesakit lain / <i>Ability to ask for a private room / sharing with less people</i>								
AC217	Keselesaan hospital (cth: kebersihan, susun atur kerusi, ruang, dll.) / <i>Comfort of hospital (e.g. cleanliness, setting of chairs, space, etc.)</i>								
AC218	Adanya ujian/ peralatan perubatan <i>Availability of investigations/ medical equipment</i>								
AC219	Adanya doktor pakar di hospital pakar <i>Availability of specialist (s) at the specialist hospital</i>								
AC220	Dibenarkan memilih doktor <i>Allowed to choose the doctor</i>								
AC221	Tempoh menunggu untuk berjumpa doktor sebaik tiba di hospital <i>The waiting time to see a doctor once arrived at the hospital</i>								
AC222	Masa yang diluangkan oleh doktor untuk pesakit <i>The amount of time the doctor spends with a patient</i>								
AC223	Kebolehan doktor memberi diagnosis dan memberi rawatan yang betul / <i>The ability of the doctor to give you the correct diagnosis and treatment</i>								
AC224	Kejelasan penerangan doktor berkenaan penyakit, ujian dan prosedur / <i>Clarity of doctor's explanation regarding the illness, test and procedure</i>								
AC225	Budi bahasa dan kesediaan doktor, penolong pegawai perubatan & jururawat untuk membantu / <i>Courtesy and helpfulness of doctor, assistant medical officer and nurse</i>								
AC226	Keberkesanan perkhidmatan / rawatan <i>The outcome of services / treatment</i>								
AC227	Caj rawatan <i>Treatment charges</i>								
AC228	Pandangan anda secara keseluruhan <i>Your overall impression</i>								

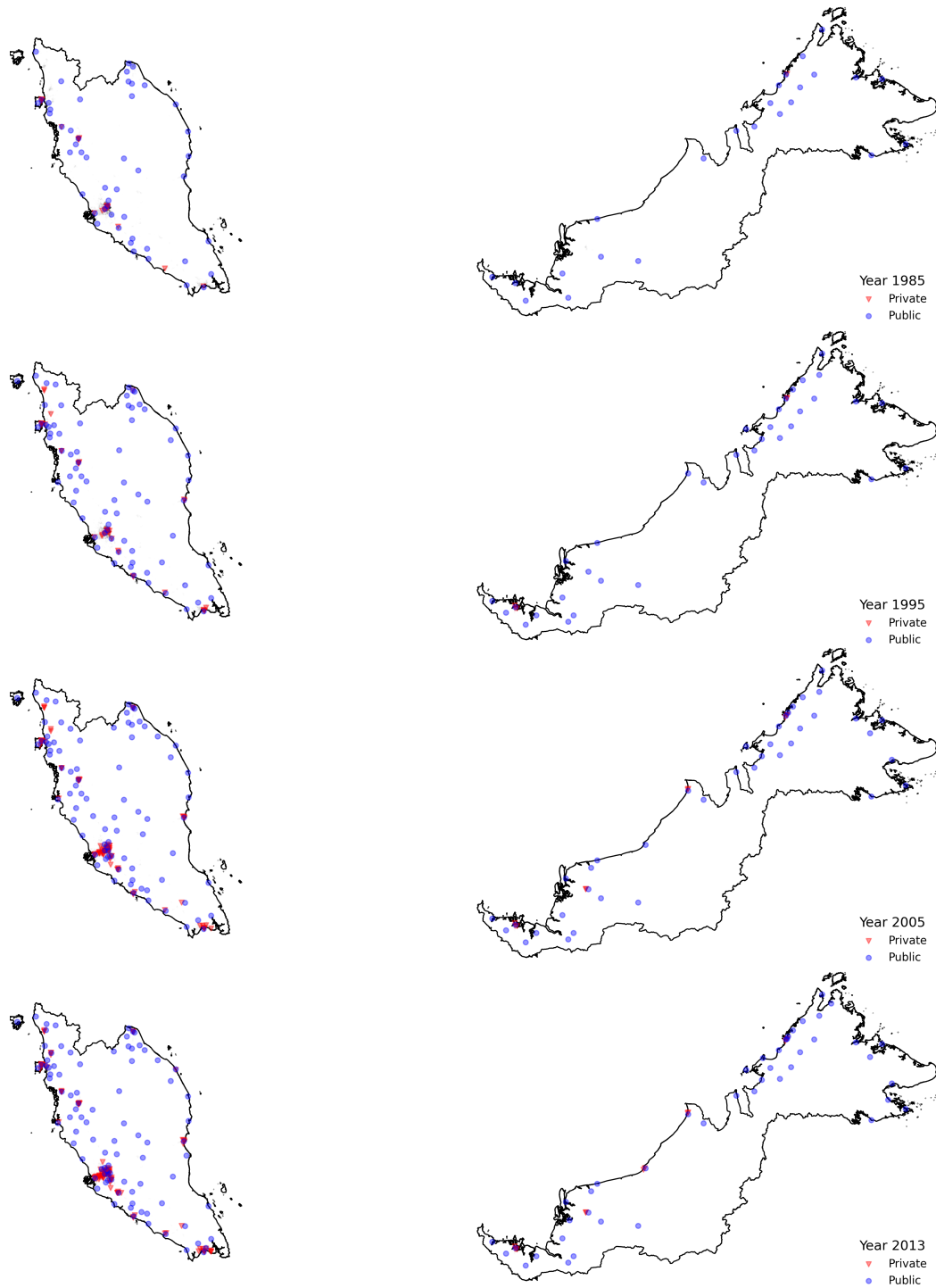
A.3 Additional Tables and Figures on Data and Context

Table A.1: Summary Statistics

	Public Hospitals		Private Hospitals
	Specialist	Non-Specialist	
A. Hospital Characteristics (2013)			
N Hospitals	61	74	134
Avg. Physician	226.39	12.36	34.89
Avg. Other Staff	719.79	80.91	133.29
Avg. Beds	509.08	88.80	94.06
Avg. Inpatient Admissions	34,291	5,432	8,395
Avg. Outpatient Visits	96,276	52,539	28,116
Avg. Bed Occupancy Rate (%)	73.9	47.33	53.88
<i>Ownership Group (%)</i>			
Government	61 (100%)	74 (100%)	-
Independent	-	-	87 (65%)
Columbia Asia	-	-	11 (8%)
KPJ	-	-	22 (16%)
Pantai	-	-	11 (8%)
Sime Darby	-	-	3 (2%)
B. Maternity Services			
Avg. Vaginal Deliveries	3,176	586	589
Avg. District Market Share (Deliveries)	0.70	0.79	0.08
Price (MYR)	100	100	3,306
	Survey Data		
	Public Hospitals		Private Hospitals
C. Survey Data			
Indv. Monthly Income (MYR)	1.52		2.54
Distance Public (km)	13.17		10.28
Distance Private (km)	31.82		15.31
Private Insurance	0.16		0.52
Chronic Disease	0.70		0.61
Quality Rating (1-5)	4.03		3.83
Wait Time Satisfaction	3.23		3.82

Notes: Panel A shows characteristics for 269 hospitals from the National Healthcare Establishment and Workforce Survey (NHEWS). Panel B presents maternity service statistics for normal vaginal deliveries from Ministry of Health electronic health records (SMRP for public, PHDD for private hospitals). Public hospital prices reflect standardized subsidized rates for third-class wards. Private hospital prices are minimum advertised rates from primary data collection (websites, social media, direct contact). Panel C shows stated preferences from 15,296 families with childbearing intentions in the National Health and Morbidity Survey (NHMS) 2015, split by hospital type preference. Distance measured as straight-line distance from households to nearest hospital within each district. Income in thousands of MYR. Quality rating and wait time satisfaction on 1-5 Likert scales.

Figure A.3: Public and Private Hospital Locations (1982-2013)



Note: Data on hospital locations are from the National Healthcare Establishment Workforce Survey (2013). This figure shows the locations of public and private hospitals in Malaysia from 1982 to 2013.

Figure A.4: Example Hospital Images

A. Public Specialist



B. Public Non-Specialist



C. Private Hospitals

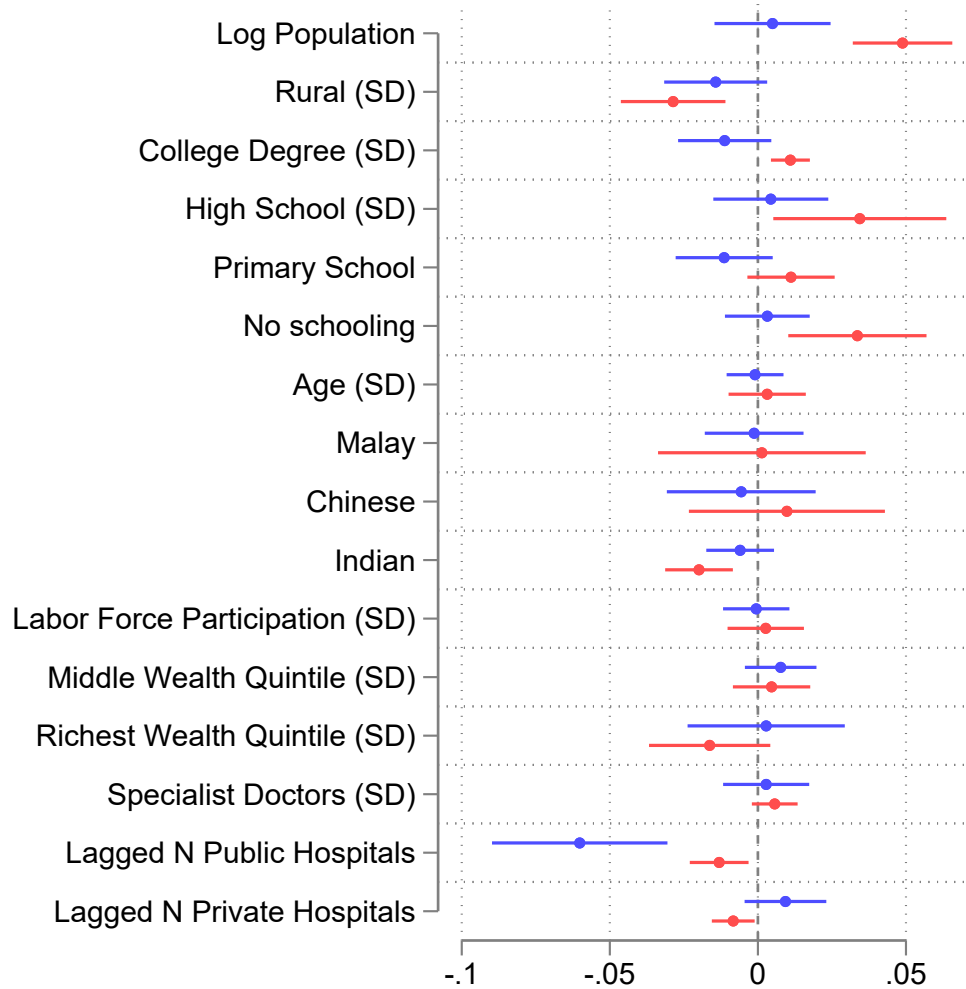


D. Maternity Centers



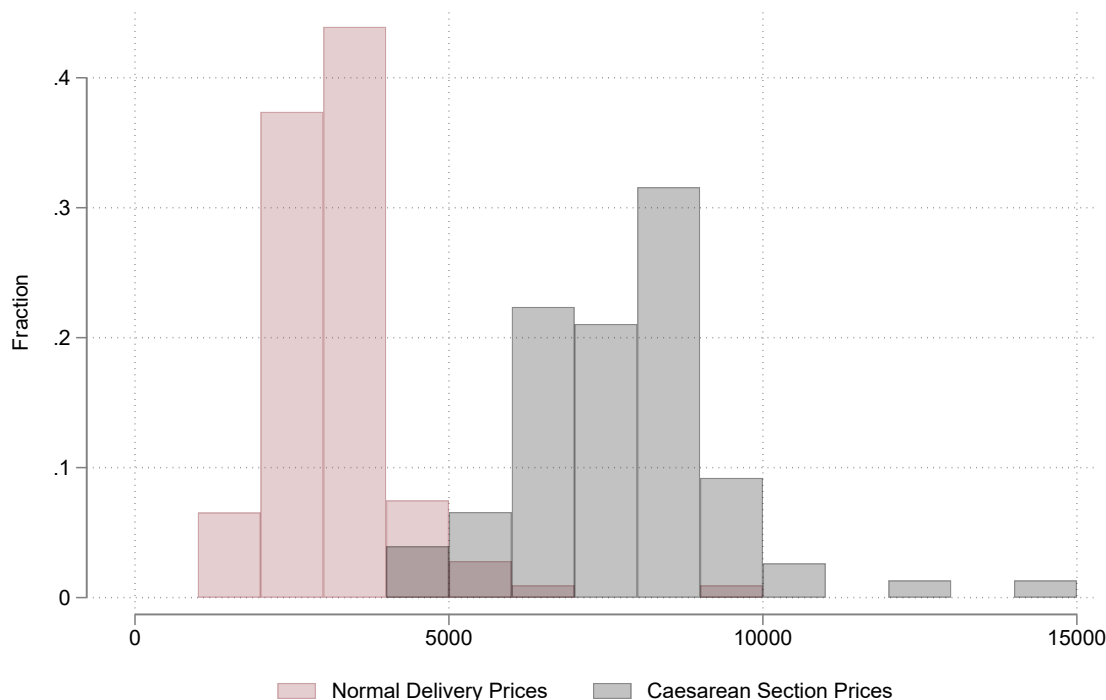
Note: These panels show examples of what hospitals in Malaysia look like based on their categories.

Figure A.5: Determinants of Public and Private Hospital Entry (Full Coefficient Set)



Note: These coefficients are average marginal effects from logit regressions with year fixed effects of public (or private) hospital entry on a set of district characteristics.

Figure A.6: Distribution of Birth Delivery Prices in Private Hospitals (MYR)



Note: This figure shows the distribution of normal (vaginal) delivery and caesarean section prices in private hospitals in Malaysia. Most maternity packages offer only normal delivery packages.

Figure A.7: Selected Excerpts from Malaysia Planning Documents

V.—CURATIVE SERVICES

544. In the field of curative medicine, measures will be taken to establish institutional facilities in areas which are still without them, to improve existing facilities and to increase the number of doctors, medical technicians, nurses and mid-wives. In Malaya major schemes in this category are mainly hospitals already approved under the previous Plan.

First Malaysia Plan 1966-1970

795. Pada ketika ini ada lebih kurang 17,000 katil di-hospital² umum dan daerah di-Malaysia Barat. Bilangan katil² di-hospital² ini bukan sahaja akan di-tambah tetapi juga kemudahan² yang terdapat di-hospital² akan juga di-perbaiki lagi. Lanakah² akan di-ambil baki menubuhkan kemudahan² perubatan di-daerah² yang tidak mempunyai-nya, memperbaiki kemudahan² yang sedia ada dan juga menambahkan bilangan doktor, kakitangan perubatan, jururawat dan bidan. Untuk menchapai tujuan² ini satu rancangan memajukan pembangunan hospital² baharu, pembesaran dan kerja² memperbaiki kemudahan² yang ada dan latehan untuk kakitangan² saperti yang di-perlukan akan di-laksanakan.

"Steps will be taken to expand access to health care in districts that lack health care access"

Second Malaysia Plan 1971-1975

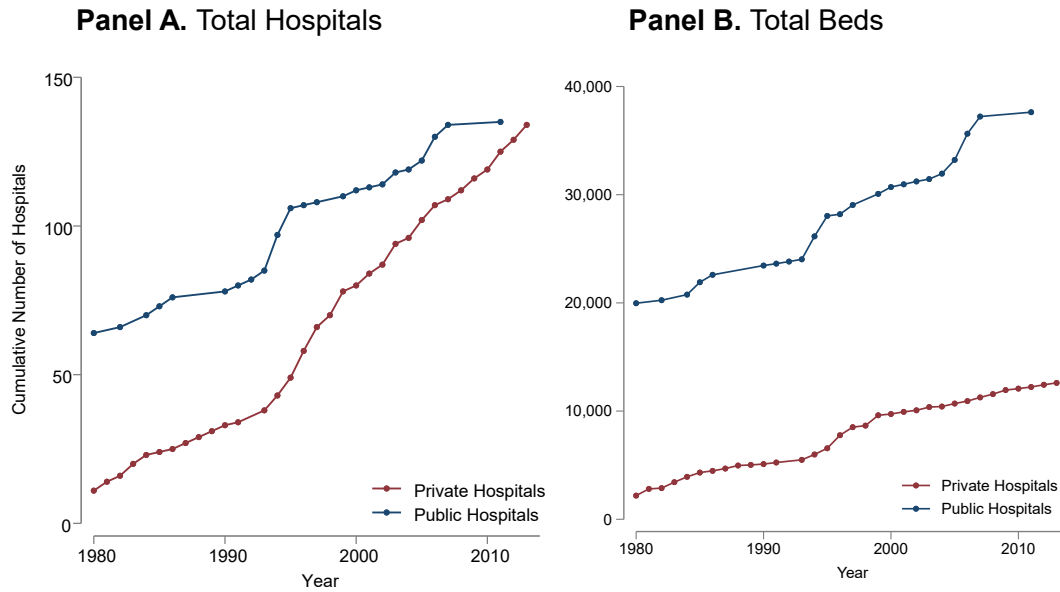
17.28 The strategies for health sector development during the Eighth Plan period will include the following :

- ☐ improving accessibility to affordable and quality healthcare;
- ☐ expanding the wellness programme;
- ☐ promoting coordination and collaboration between public and private sector providers of health care;
- ☐ increasing the supply of various categories of health manpower;
- ☐ strengthening the telehealth system to promote Malaysia as a regional centre for health services;
- ☐ enhancing research capacity and capability of the health sector;
- ☐ developing and instituting a healthcare financing scheme; and
- ☐ strengthening the regulatory and enforcement functions to administer the health sector, including traditional practitioners and medical products.

Eighth Malaysia Plan 2001-2005

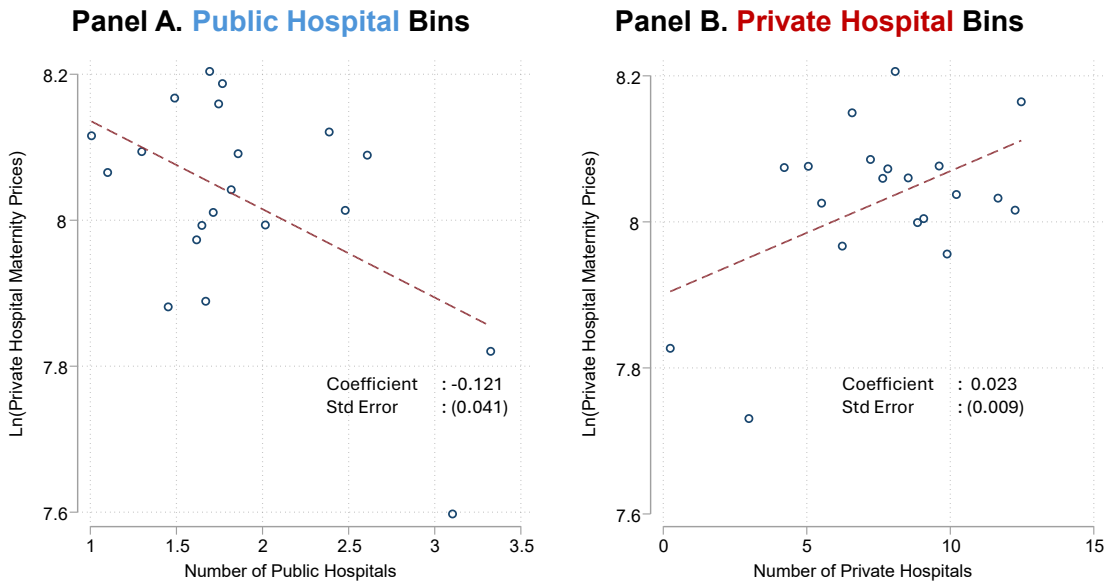
Note: These excerpts are taken from various planning documents related to healthcare development in Malaysia. These panels show the commitment of the Malaysian government in prioritizing access to healthcare.

Figure A.8: Total Count and Beds by Public and Private Hospitals



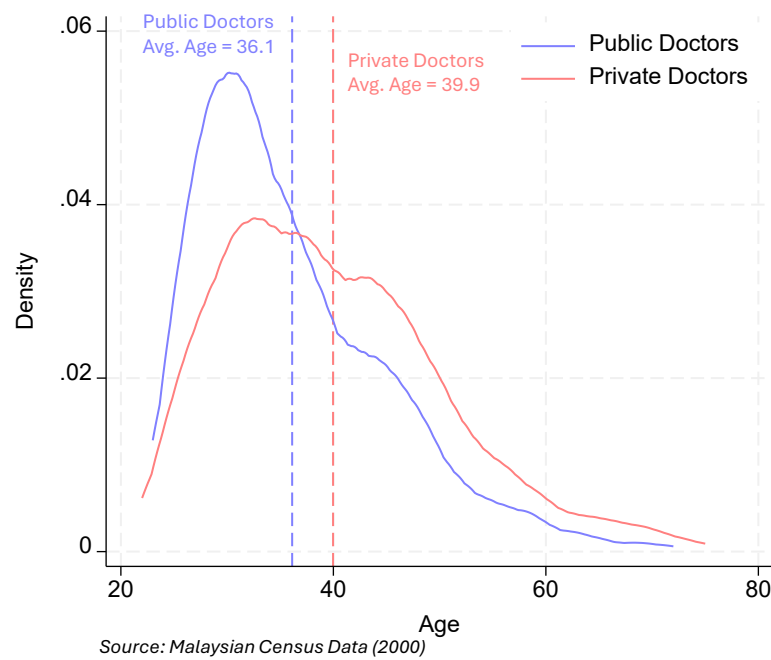
Note: This figure shows the total count of hospitals and the number of beds available in public and private hospitals in Malaysia between 1980 and 2014.

Figure A.9: Private Hospital Normal Delivery Prices by Number of Public/Private Hospitals in District



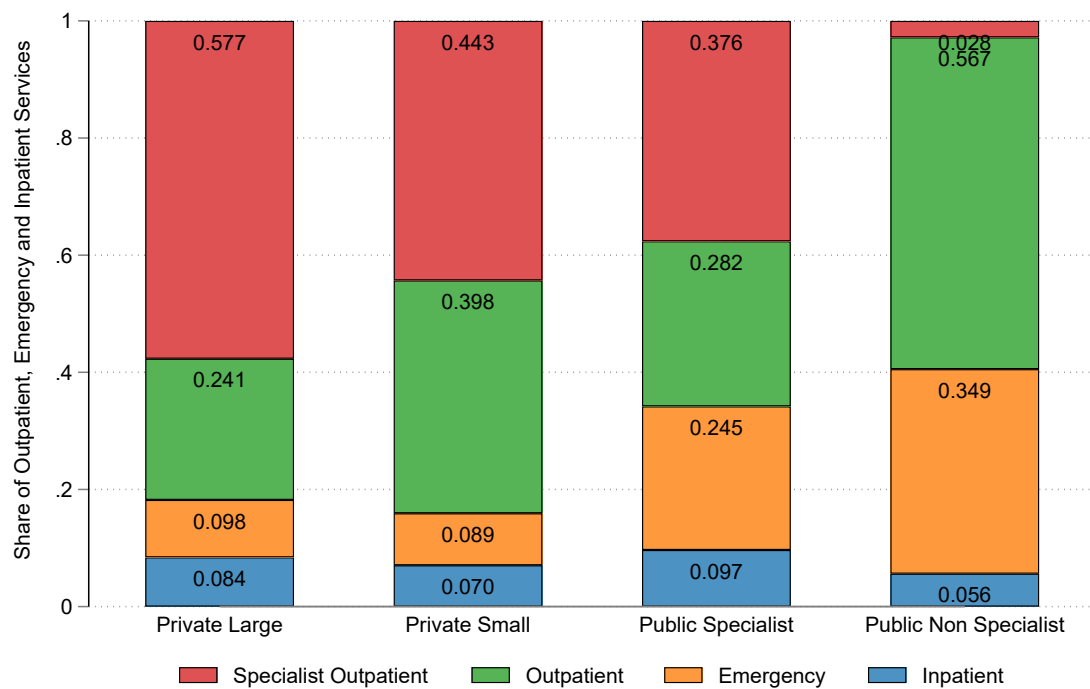
Note: Panel A figure shows a binscatter of private hospital normal delivery prices against the number of public hospitals within the same district. Panel B shows a binscatter against the number of private hospitals within the same district. This figure shows descriptive evidence on public competitive pressures on private pricing.

Figure A.10: Physician Average Age by Public & Private Hospitals



Note: This figure shows the average age of physicians working in public and private hospitals in Malaysia. The difference is approximately 3.8 years. This roughly coincides with the four-year mandatory service obligation in the public sector.

Figure A.11: Proportion of Outpatient, Emergency and Inpatient Visits at Private and Public Hospitals

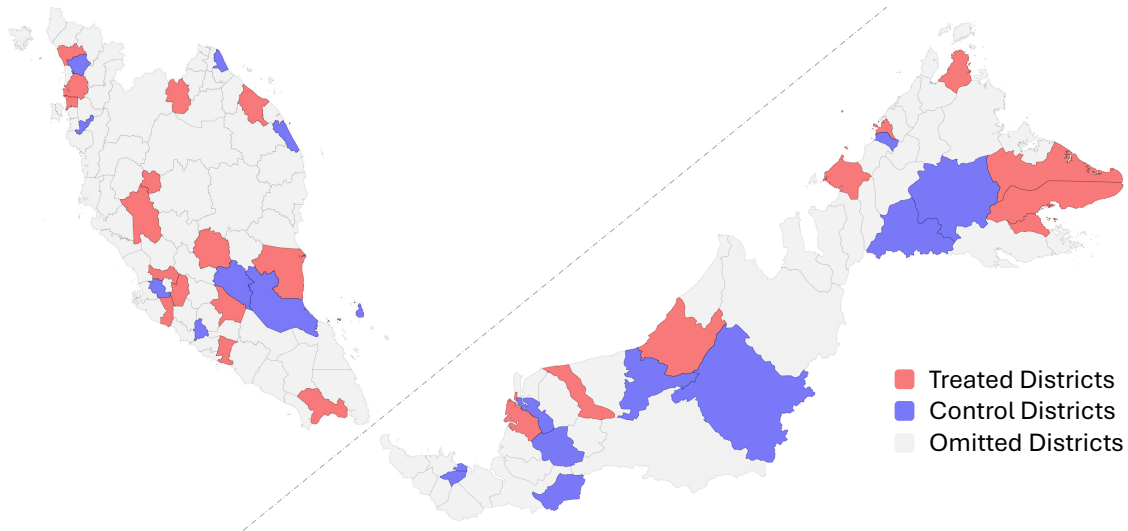


Note: Data from the National Healthcare Establishment Workforce Survey (2013). Panels show the distribution of outpatient, emergency, and inpatient visits across public and private hospitals.

B Further Details on Reduced Form

B.1 Balance and Sample

Figure B.1: Sample of Districts in Event Studies



Note: Red districts are included in the event study design as treated districts, while blue districts are controls. Grey districts are omitted from the event study.

Table B.1: Pre-Treatment Summary Statistics by Treatment Status in 1991

Variable	Treated	Never Treated	Difference	P-value
<i>Panel A. Demographics</i>				
Log Population	11.754 (1.115)	11.057 (1.106)	0.697	0.037
Rural Population Share	0.602 (0.310)	0.782 (0.312)	-0.180	0.063
Average Age	23.717 (2.043)	25.258 (2.536)	-1.541	0.031
Female Share	0.482 (0.024)	0.497 (0.039)	-0.015	0.123
Chinese Share	0.215 (0.151)	0.200 (0.194)	0.015	0.773
Malay Share	0.416 (0.306)	0.447 (0.354)	-0.031	0.752
Indian Share	0.071 (0.080)	0.047 (0.063)	0.024	0.281
Labor Force Participation	0.632 (0.058)	0.642 (0.104)	-0.010	0.669
<i>Panel B. Socioeconomic Status</i>				
Poorest	0.434 (0.260)	0.494 (0.286)	-0.060	0.476
Middle	0.307 (0.129)	0.293 (0.153)	0.014	0.733
Richest	0.258 (0.181)	0.213 (0.205)	0.045	0.446
<i>Panel C. Education</i>				
College/University	0.021 (0.020)	0.021 (0.033)	0.000	0.968
Secondary Completed	0.226 (0.080)	0.222 (0.079)	0.004	0.867
Primary Completed	0.201 (0.041)	0.194 (0.041)	0.007	0.617
Some Primary Education	0.201 (0.030)	0.204 (0.034)	-0.003	0.766
<i>Panel D. Health Facilities</i>				
Dist. to Pub Hosp (km)	36.522 (28.627)	41.423 (40.336)	-4.901	0.630
Dist. to Pri Hosp (km)	102.979 (105.322)	106.691 (112.125)	-3.712	0.907
N Public Hospitals	0.320 (0.557)	0.182 (0.395)	0.138	0.338
N Private Hospitals	0.680 (1.725)	0.273 (0.935)	0.407	0.329
N Specialist Physicians	0.050 (0.111)	0.018 (0.058)	0.032	0.266

Notes: This table compares the 25 treatment districts with the 22 never treated districts based on pre-treatment characteristics in 1991. Standard deviations in parentheses. Unit of observation is districts. All data from the 1991 Malaysian Census and hospital panel data. Distances are straight-line kilometers to facilities from 1km grids, collapsed at the district level. Wealth quintiles are constructed from household assets (electricity, water supply, telephone, automobiles, air conditioning, washing machine, refrigerator, television, VCR, radio, toilet, wall material).

B.2 Full Event Study Coefficients

[Table B.2](#) tabulates the full set of event study coefficients from [Equation 11](#).

Table B.2: Effects of Public Hospital Construction on Private Hospitals (Full Coefficient Table)

Time since Construction	(1) All Public Hospitals	(2) Specialist Public Hospitals	(3) Non-Specialist Public Hospitals
-10	-0.314 (0.041)	-1.190 (0.293)	0.415 (0.087)
-9	-0.075 (0.073)	-0.527 (0.215)	0.315 (0.071)
-8	-0.059 (0.069)	-0.329 (0.237)	0.214 (0.068)
-7	-0.012 (0.077)	-0.310 (0.228)	0.224 (0.060)
-6	-0.058 (0.067)	-0.286 (0.211)	0.150 (0.044)
-5	-0.030 (0.048)	-0.166 (0.144)	0.109 (0.043)
-4	-0.048 (0.048)	-0.130 (0.132)	0.055 (0.041)
-3	-0.035 (0.036)	-0.090 (0.084)	0.031 (0.043)
-2	-0.047 (0.027)	-0.077 (0.084)	-0.012 (0.040)
0	0.082 (0.048)	0.200 (0.095)	-0.070 (0.034)
1	0.094 (0.051)	0.244 (0.107)	-0.101 (0.037)
2	0.142 (0.056)	0.355 (0.135)	-0.132 (0.032)
3	0.127 (0.058)	0.332 (0.140)	-0.134 (0.032)
4	0.155 (0.066)	0.387 (0.167)	-0.145 (0.032)
5	0.171 (0.078)	0.436 (0.198)	-0.169 (0.038)
6	0.344 (0.123)	0.766 (0.261)	-0.198 (0.039)
7	0.421 (0.148)	0.861 (0.326)	-0.169 (0.016)
8	0.179 (0.195)	0.392 (0.317)	-0.209 (0.024)
9	0.213 (0.223)	0.445 (0.345)	-0.261 (0.030)
10+	0.323 (0.313)	0.673 (0.481)	-0.295 (0.020)
Observations	846	648	594
R-squared	0.965	0.970	0.946

Standard errors in parentheses, clustered at the district level.

B.3 Alternative Inference Procedures

With 14 specialist treatment events, 11 non-specialist treatment events, and 22 never-treated control districts, conventional clustered standard errors may overstate statistical significance. In this appendix section, I implement two alternative inference procedures that do not rely on asymptotic approximations.

Pairs Cluster Bootstrap. Standard clustered standard errors assume that as the number of clusters grows large, the sampling distribution of the estimator converges to a normal distribution. With as few as 33 clusters in the non-specialist sample, this approximation may be poor. The pairs cluster bootstrap directly estimates this variability by repeatedly resampling the data and computing the statistic each time, rather than relying on the asymptotic formula.

Specifically, I resample entire districts (not individual observations) with replacement ($B = 999$), preserving the within-district correlation structure, and re-estimate the Sun and Abraham (2021) specification on each draw (Cameron et al., 2008). Each bootstrap draw produces a new estimate of the average post-treatment effect. The standard deviation of these 999 estimates is the bootstrap standard error. The two-sided percentile p -value is twice the share of bootstrap estimates that fall on the opposite side of zero from the original coefficient, and the 95 percent confidence interval is constructed from the 2.5th and 97.5th percentiles of the bootstrap distribution.

Table B.3 tabulates the results. For the specialist result, the bootstrap standard error is approximately three and a half times the analytic standard error (0.381 versus 0.108), with a two-sided percentile p -value of 0.042 and a 95 percent confidence interval of [0.013, 1.504]. The specialist result therefore remains statistically distinguishable from zero at the five percent level even under this more conservat-

ive procedure. The bootstrap standard error for the non-specialist result inflates by a factor of seventeen (0.161 versus 0.009). The 95 percent percentile confidence interval is $[-0.582, 0.000]$ and the two-sided bootstrap p -value is 0.098, consistent with directional crowd-out that the few-cluster correction does not estimate precisely. The pooled estimate is also statistically indistinguishable from zero under the corrected inference (95 percent CI: $[-0.192, 1.124]$), consistent with the opposing-sign type-specific effects canceling in the pooled average.

Randomization Inference. Randomization inference asks a different question than the bootstrap. Rather than asking whether the standard errors are correctly sized, it asks: could an effect this large have arisen by chance, simply because the specialist hospital events across 14 treated districts happened to be places where private hospitals were growing for reasons unrelated to public hospital construction? The test answers this by asking what the estimated effect would look like if treatment had been randomly assigned to a different set of districts.

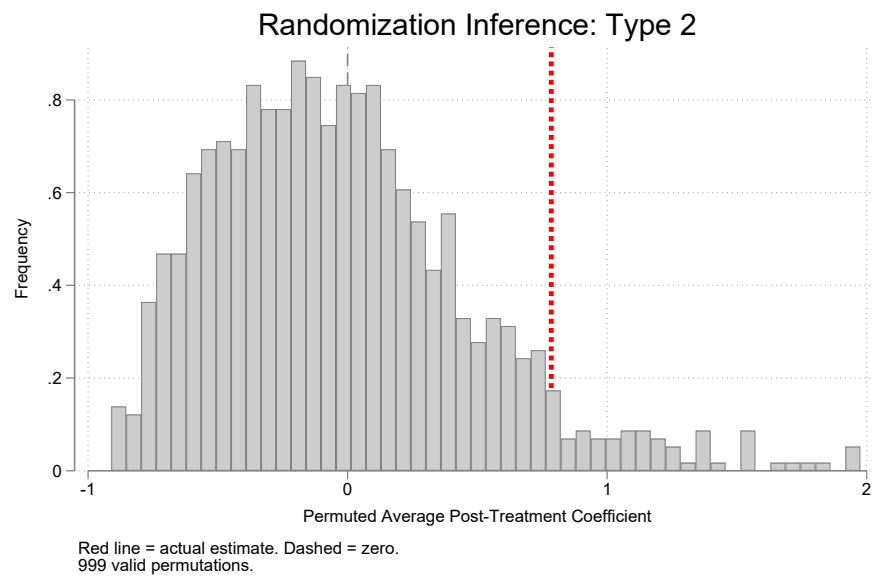
Under the null hypothesis of no treatment effect, I randomly select 14 of the 36 total districts (14 treated plus 22 never-treated), assign them the original vector of treatment years, and designate the remaining 22 as never-treated controls (Young, 2019). I then re-estimate the Sun-Abraham specification and store the average post-treatment coefficient. Repeating this 999 times generates a distribution of placebo effects under the null. If the actual estimate is unusually large relative to this distribution, treatment assignment matters. [Figure B.2](#) shows the result. The permutation distribution is centered near zero, showing the test has power against the null. The actual specialist estimate of 0.785 lies in the right tail, with a one-sided p -value of 0.06 and a two-sided p -value of 0.08.

Table B.3: Alternative Inference for Average Post-Treatment Effects

	All	Specialist	Non-specialist
	(1)	(2)	(3)
<i>Panel A. Main Table Result (Clustered)</i>			
Coefficient	0.465	0.785	−0.171
Standard error	(0.094)	(0.108)	(0.009)
<i>Panel B. Pairs Cluster Bootstrap (B = 999)</i>			
Bootstrap SE	0.333	0.381	0.161
SE inflation	3.6×	3.5×	17.1×
Percentile <i>p</i> -value (two-sided)	0.180	0.042	0.098
95% CI	[−0.192, 1.124]	[0.013, 1.504]	[−0.582, 0.000]
<i>Panel C. Randomization Inference (B = 999)</i>			
One-sided <i>p</i> -value	—	0.060	—
Two-sided <i>p</i> -value	—	0.078	—

Notes: Panel A reports analytic standard errors clustered at the district level from [Table 1](#). Panel B reports pairs cluster bootstrap results that resample entire districts with replacement. Panel C reports randomization inference *p*-values from permuting treatment status across all 36 districts (14 treated, 22 never-treated), preserving the original treatment year vector. RI is reported only for the specialist specification (Column 2), which is the paper’s main result.

Figure B.2: Randomization Inference Distribution for Specialist Hospital Effects



Notes: Distribution of average post-treatment coefficients from 999 permutations that randomly assign 14 of 36 districts to treatment status. The red vertical line marks the actual estimate (0.785). The dashed line marks zero. The permutation distribution is centered near zero (mean = -0.026), showing the test has power against the null.

B.4 Physician Supply: Lag Robustness

The physician supply mechanism occurs with a delay: residency training in Malaysia requires four to six years. The main physician results in [Table 3](#) use lag 3 to capture accumulated effects over a training cycle. [Table B.4](#) reports the effects across lag lengths 0 through 5. For specialist hospitals (Panel A), the effect grows monotonically from 0.547 at lag 0 to 1.081 at lag 3, where it stabilizes. This pattern is consistent with the training timeline: the contemporaneous jump (lag 0) reflects the arrival of senior specialists at the district, while the growth through lags 1–3 captures the first cohorts completing residency training and entering private practice. The stabilization at lag 3 suggests the pipeline reaches a steady state within approximately one training cycle. For non-specialist hospitals (Panel B), the effects are close to zero and statistically insignificant at all lags, showing that the physician supply channel is specific to hospitals with training programs.

Table B.4: Robustness: Effects of Public Hospitals on Private Specialists by Lag Length

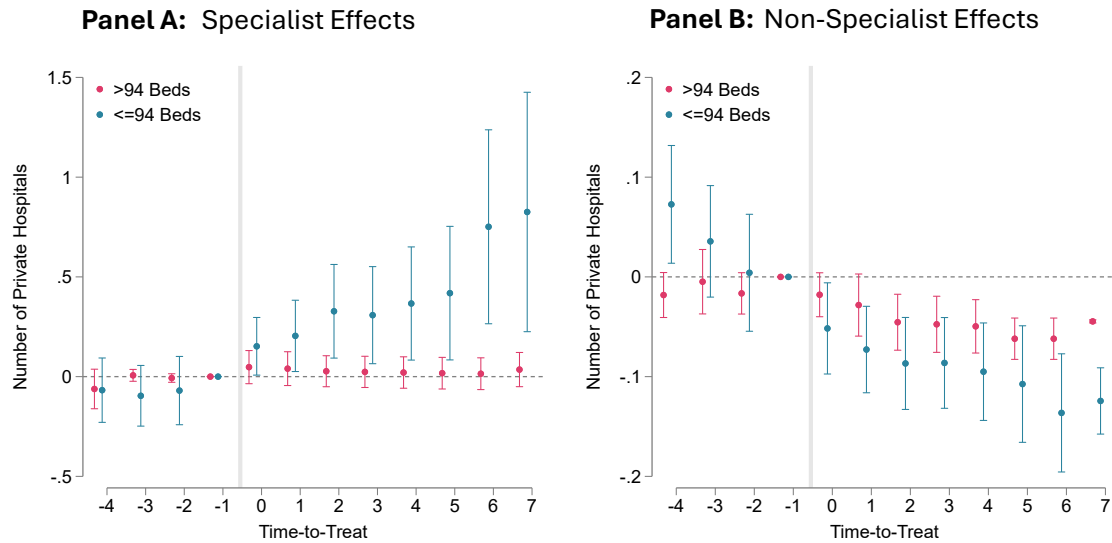
	Private Specialist Physicians (in 100s)					
	Lag 0	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5
Panel A: Specialist Public Hospitals						
Number of Hospitals	0.547 (0.302)	0.772 (0.477)	0.743 (0.391)	1.081 (0.471)	1.081 (0.471)	1.081 (0.471)
Observations	58	58	60	62	62	62
Mean Dep. Var.	0.309	0.378	0.375	0.411	0.411	0.411
R^2	0.872	0.891	0.891	0.858	0.858	0.858
Panel B: Non-Specialist Public Hospitals						
Number of Hospitals	0.063 (0.158)	0.063 (0.158)	-0.084 (0.063)	-0.084 (0.063)	-0.088 (0.066)	-0.093 (0.074)
Observations	68	68	66	66	68	68
Mean Dep. Var.	0.159	0.159	0.141	0.141	0.137	0.137
R^2	0.820	0.820	0.880	0.880	0.881	0.881

Notes: Each column presents stacked difference-in-differences estimates with different lag structures. The main physician supply results (Table 3) use lag 3. Effects for specialist hospitals strengthen at longer lags, consistent with residency training programs requiring several years to graduate specialists who then enter private practice. Panel A: specialist public hospitals. Panel B: non-specialist public hospitals. All specifications include district-by-stack and year-by-stack fixed effects. Standard errors clustered at the district level in parentheses.

B.5 Heterogeneous Effects by Hospital Size: Event Study Dynamics

Figure B.3 presents the same event study specification but splitting outcomes to small and large private hospitals. The physician supply mechanism predicts that crowd-in should concentrate among smaller private hospitals, which require fewer specialists to reach minimum viable scale and are more likely to be founded by individual physicians transitioning from public practice. Panel A shows this prediction: small private hospitals (fewer than 94 beds) show a clear positive trend following specialist public hospital construction, with effects growing steadily over the post-treatment window. Large private hospitals show essentially flat effects, consistent with large urban facilities having access to deeper physician labor markets that are less affected by any single public hospital opening. Panel B shows that non-specialist public hospitals produce weakly negative effects for both size categories, with no evidence of differential crowd-out by hospital size. The absence of size heterogeneity in the non-specialist case is consistent with demand competition operating independently of hospital scale, since a new public hospital draws patients regardless of whether the competing private facility is small or large.

Figure B.3: Effects of Specialist and Non-Specialist Public Hospitals on Small and Large Private Hospitals



Note: This figure presents four pre-period and seven post-period event study estimates from Equation 11. Panel A shows the effects of 14 new specialist public hospitals on the number of small (fewer than 94 beds) and large (94 or more beds) private hospitals within the same district. Panel B shows the effects of 11 non-specialist public hospitals. Each dot represents a point estimate with the corresponding 95 percent confidence interval shown as vertical lines. The reference period is $\ell = -1$. Standard errors are clustered at the district level.

B.6 Robustness to Violations of Parallel Trends

I assess the sensitivity of the event study estimates to violations of parallel trends (Rambachan and Roth, 2023). Rather than imposing that parallel trends holds exactly, I bound the post-treatment violation between consecutive years by $\bar{M}$ times the largest pre-treatment violation, the relative magnitudes restriction $\Delta^{RM}(\bar{M})$. I apply the procedure to the estimated coefficients from Table 1.

Table B.5 reports robust confidence sets for the fifteen-year and eight-year post-treatment windows. For the specialist treatment, the averaged effect over fifteen post-treatment years remains distinguishable from zero for violations up to a breakdown value of $\bar{M}^*$ between 0.25 and 0.30. This implies that the estimate survives post-treatment violations up to roughly 30 percent of the largest pre-treatment violation. The pooled effect is more robust, with a breakdown between 0.40 and 0.45, as its pre-treatment coefficients are smaller and more precisely estimated. Overall, moderate violations of parallel trends cannot account for the specialist result, but violations as large as the worst pre-treatment deviation could. This sensitivity analysis should be read in context of other robustness checks.

Table B.5: Sensitivity to Violations of Parallel Trends

Treatment	Window	Estimate	Original CI	$\bar{M} = 0.25$	$\bar{M} = 0.5$	$\bar{M}^*$
Specialist	15 years	0.785	[0.574, 0.995]	[0.101, 1.467]	[-0.523, 2.092]	(0.25, 0.30]
Specialist	8 years	0.448	[0.313, 0.582]	[0.073, 0.821]	[-0.259, 1.155]	(0.30, 0.35]
All public	15 years	0.465	[0.281, 0.648]	[0.144, 0.786]	[-0.077, 1.007]	(0.40, 0.45]
All public	8 years	0.192	[0.133, 0.251]	[0.047, 0.337]	[-0.077, 0.462]	(0.30, 0.35]

Notes: Robust 95% confidence sets under $\Delta^{RM}(\bar{M})$ computed with Rambachan and Roth (2023), using three pre-treatment coefficients. The estimand is the averaged effect over the first 8 or 15 post-treatment years. $\bar{M}^*$ is the breakdown bracket on a grid of step 0.05, reported as the interval between the last grid point at which zero is excluded and the first at which it is included.

B.7 Heterogeneous Effects by Institutional Context: Pre-1996 Corporatization Era

During the early 1990s, the Malaysian government pursued corporatization of public hospitals, proposing to maintain government ownership while operating them as profit-maximizing entities. This policy began with incremental reforms including corporatization of Hospital Kuala Lumpur's cardiac unit in 1992 and contracting out of drug distribution systems in 1994. The policy environment reversed with the 7th Malaysia Plan (1996–2000), when the government recommitted to purely public healthcare provision.

Table B.6 presents event study estimates for the pre-1996 period. All types of public hospitals crowded out private entry during this period, including specialist hospitals. The reversal from uniform crowd-out to heterogeneous effects after 1996 suggests that institutional context matters: crowd-in emerges specifically when public hospitals operate as purely public institutions that train specialists who subsequently enter private practice. During the corporatization era, policy uncertainty about the future structure of public hospitals and the possibility that corporatized facilities would retain trained specialists may have weakened the training pipeline that drives the post-1996 crowd-in result.

Table B.6: Effects of Public Hospital Entry on Private Hospitals: Pre-1996 Period

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	−0.084 (0.007)		
Specialist public hospitals		−0.035 (0.019)	
Non-specialist public hospitals			−0.095 (0.009)
Mean Dep. Var.	0.166	0.204	0.181
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N Districts × Year	1,024	592	784
R^2	0.886	0.943	0.900
Events	42	15	27
Estimator	SA	SA	SA

Notes: Each column presents results from separate regressions for the pre-1996 period when public hospitals faced corporatization pressures. The dependent variable is the number of private hospitals in a district. Standard errors clustered at the district level in parentheses.

B.8 Additional Robustness Tables

This subsection collects robustness checks that address specific threats to the main event study estimates.

Sun-Abraham versus TWFE. A concern with two-way fixed effects estimation under staggered treatment timing is that already-treated units can serve as implicit controls for later-treated units, biasing the estimated treatment effect when effects vary across cohorts (Sun and Abraham, 2021). The Sun-Abraham interaction-weighted estimator addresses this by separately estimating cohort-specific effects and aggregating them. If treatment effect heterogeneity across cohorts is minimal, the two estimators should produce similar results. [Table B.7](#) shows this.

Balancing Regressions. If public hospitals are systematically built in districts that are also experiencing increased private hospital entry, the event study could confound treatment effects with pre-existing trends correlated with district characteristics. To test this, I first predict the number of private hospitals using the district-level covariates from [Table B.1](#), then estimate [Equation 11](#) with this predicted outcome as the dependent variable. [Figure B.4](#) shows that the correlation between public hospital construction and predicted private entry disappears once district fixed effects are included, indicating that the event study design effectively controls for the observable confounders.

Excluding Multiple-Treatment Districts. Three districts received more than one public hospital during the study period. If entry of a second public hospital is endogenous to the private hospital response to the first, including these districts could bias the estimates. [Table B.8](#) drops these three districts. The specialist estimate is somewhat larger in the restricted sample (0.937 versus 0.785), likely reflecting

composition as the three excluded districts may have experienced below-average crowd-in. The non-specialist estimate is unchanged (-0.171). Neither change suggests the results are driven by districts with multiple treatments.

Later-Treated as Controls. The main specification uses never-treated districts as the control group. An alternative is to use the later-treated cohort as controls, which ensures that both treatment and control groups eventually receive a public hospital and may therefore be more comparable in terms of unobservable characteristics that predict hospital construction. [Table B.9](#) implements this approach. The specialist estimate of 0.599 is somewhat attenuated relative to the main estimate (0.785) but remains economically large and positive. The non-specialist specification cannot be estimated under this design because both the later-treated control districts and the non-specialist treated districts had zero private hospital entrants.

Comparison Across Staggered DiD Estimators. The recent literature has proposed several alternative estimators for staggered difference-in-differences designs, each making different assumptions about treatment effect dynamics and the appropriate control group. [Table B.10](#) reports the average post-treatment effect from four estimators: Sun-Abraham, Borusyak-Jaravel-Spiess, Callaway-Sant’Anna, and de Chaisemartin-D’Haultfœuille. All four estimators agree on the qualitative pattern: specialist hospitals crowd in and non-specialist hospitals produce null effects. The Callaway-Sant’Anna estimator produces larger point estimates (1.424 for specialist). All four estimators use never-treated districts as controls, so the differences in point estimates reflect differences in how each estimator aggregates cohort-specific effects and handles treatment effect dynamics. Standard errors also vary, reflecting differences in how each accounts for treatment effect heterogeneity and serial correlation.

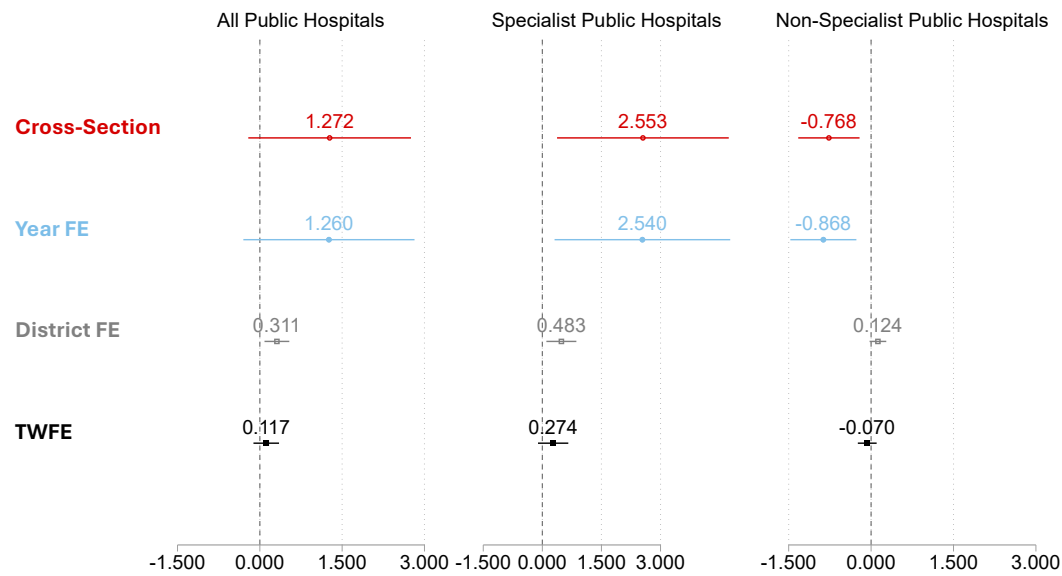
Bed-Size Weighted Effects. If the physician training spillover scales with hospital capacity, larger specialist hospitals should generate proportionally greater crowd-in. [Table B.11](#) tests this by weighting each treatment event by the entering public hospital’s bed capacity. The bed-weighted specialist effect (1.069) exceeds the unweighted effect (0.785), consistent with training capacity scaling with hospital size. The non-specialist effect is essentially unchanged under weighting (−0.161 versus −0.171), as expected if crowd-out operates through demand competition rather than a size-dependent mechanism.

Table B.7: Sun-Abraham vs. Two-Way Fixed Effects Estimates

	Number of Private Hospitals		
	All	Specialist	Non-specialist
<i>Panel A. Sun-Abraham</i>			
Post-treatment average	0.465 (0.094)	0.785 (0.108)	−0.171 (0.009)
<i>Panel B. TWFE with endpoint binning</i>			
Post-treatment average	0.477 (0.097)	0.791 (0.111)	−0.144 (0.010)
Difference (SA − TWFE)	−0.012	−0.006	−0.027
% difference	2.5%	0.8%	15.8%

Notes: Panel A reproduces the main Sun-Abraham interaction-weighted estimates from [Table 1](#). Panel B reports conventional TWFE estimates with endpoint binning (i.e., binning the earliest and latest relative time indicators to ensure all event times are estimable). The near-identical coefficients for specialist hospitals (less than 1 percent difference) indicate that treatment effect heterogeneity across cohorts is negligible. Standard errors clustered at the district level in parentheses.

Figure B.4: Balancing Regressions



Notes: I first predict the number of private hospitals using district-level characteristics in [Table B.1](#). I then estimate [Equation 11](#) with this predicted outcome as the dependent variable under alternative fixed effects structures. The correlation between public hospital construction and predicted private entry disappears once district fixed effects are included, suggesting that the event study design effectively controls for observable confounders.

Table B.8: Main Effects Robustness: Dropping Multiple Treated Districts

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.457 (0.151)		
Specialist public hospitals		0.937 (0.114)	
Non-specialist public hospitals			−0.171 (0.009)
Mean Dep. Var.	1.282	1.709	0.631
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
N Districts × Year	792	594	594
R^2	0.953	0.957	0.930
Events	22	11	11
Estimator	SA	SA	SA

Notes: Excludes three districts that received multiple public hospitals during 1996–2013. Standard errors clustered at the district level in parentheses.

Table B.9: Main Effects Robustness: Later-Treated as Control

	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.684 (0.081)		
Specialist public hospitals		0.599 (0.088)	
Non-specialist public hospitals			—
Mean Dep. Var.	0.979	1.278	—
District Fixed Effects	Yes	Yes	—
Year Fixed Effects	Yes	Yes	—
R^2	0.980	0.975	—
Events	25	14	—
Estimator	SA	SA	—

Notes: Uses later-treated districts as controls. Column (3) cannot be estimated because both later-treated control districts and non-specialist treated districts had zero private hospital entrants. Standard errors clustered at the district level in parentheses.

Table B.10: Post-Treatment Effects on Private Hospital Entry: Comparison of Estimators

	SA (1)	BJSp (2)	CS (3)	DdH (4)
All public hospitals	0.465 (0.094)	0.296 (0.267)	1.153 (0.323)	0.289 (0.177)
Specialist public hospitals	0.785 (0.108)	0.659 (0.292)	1.424 (0.336)	0.558 (0.191)
Non-specialist public hospitals	−0.171 (0.009)	−0.281 (0.236)	−0.208 (0.181)	−0.119 (0.110)

Notes: Each column presents post-treatment average effects using different estimators for staggered DiD designs. SA: Sun and Abraham (2021). BJSp: Borusyak et al. (2024). CS: Callaway and Sant’Anna (2021). DdH: de Chaisemartin and D’Haultfœuille (2024). All specifications use never-treated districts as controls. Differences in point estimates reflect variation in how each estimator aggregates cohort-specific effects and handles treatment effect dynamics. Standard errors in parentheses.

Table B.11: Bed-Size Weighted Event Study Estimates

	Unweighted			Bed-Size Weighted		
	(1)	(2)	(3)	(4)	(5)	(6)
All public hospitals	0.465 (0.094)			1.013 (0.109)		
Specialist		0.785 (0.108)			1.069 (0.111)	
Non-specialist			−0.171 (0.009)			−0.161 (0.014)
Mean Dep. Var.	1.303	1.701	0.631	1.303	1.701	0.631
Pre-Treatment Mean	1.440	2.571	0.000	1.440	2.571	0.000
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N Districts × Year	846	648	594	846	648	594
Events	25	14	11	25	14	11
Estimator	SA	SA	SA	SA	SA	SA

Notes: Columns 1–3 reproduce the main (unweighted) event study results from Table 1. Columns 4–6 weight each treatment event by the entering public hospital’s bed capacity. The bed-weighted specialist effect (1.069) exceeds the unweighted effect (0.785), indicating that larger specialist hospitals generate proportionally greater crowd-in, consistent with training capacity scaling with hospital size. The non-specialist effect is essentially unchanged, as expected if crowd-out operates through demand competition rather than a size-dependent mechanism. Standard errors clustered at the district level in parentheses.

B.9 Synthetic Difference-in-Differences

To address concerns about pre-treatment imbalances between treatment and control districts, I re-estimate the main results using the synthetic difference-in-differences estimator (Arkhangelsky et al., 2021). This method constructs synthetic control units that optimally weight both untreated districts and pre-treatment time periods to better match the treated units' characteristics and trends. The specialist SDID estimate (0.692) is positive and similar in magnitude to the main SA estimate (0.785). The non-specialist SDID estimate is attenuated and statistically insignificant, likely reflecting the difficulty of constructing well-matched synthetic controls with only 11 non-specialist treatment events in predominantly rural districts with limited pre-treatment private hospital variation.

Table B.12: Average Treatment Effects on Private Entrants — Synthetic Difference-in-Differences

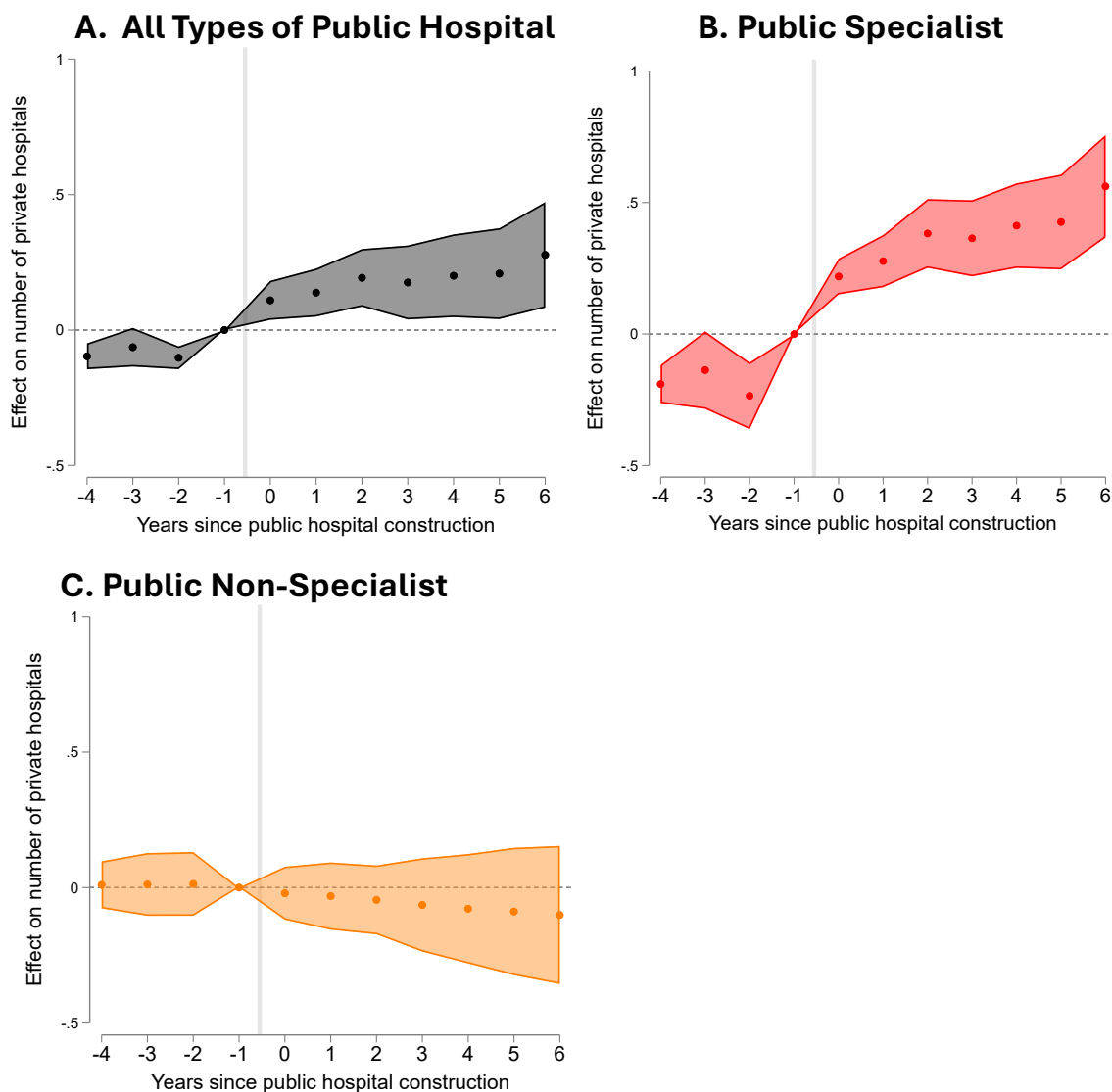
	Number of Private Hospitals		
	(1)	(2)	(3)
All public hospitals	0.406 (0.168)		
Specialist public hospitals		0.692 (0.275)	
Non-specialist public hospitals			−0.016 (0.030)
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Estimator	SDID	SDID	SDID

Notes: Synthetic DiD estimator of Arkhangelsky et al. (2021). Standard errors in parentheses.

B.10 Matching

I use coarsened exact matching (CEM) to balance treatment and control districts on rurality, which shows a significant pre-treatment imbalance, and the number of

Figure B.5: Synthetic Difference-in-Differences: Dynamic Treatment Effects

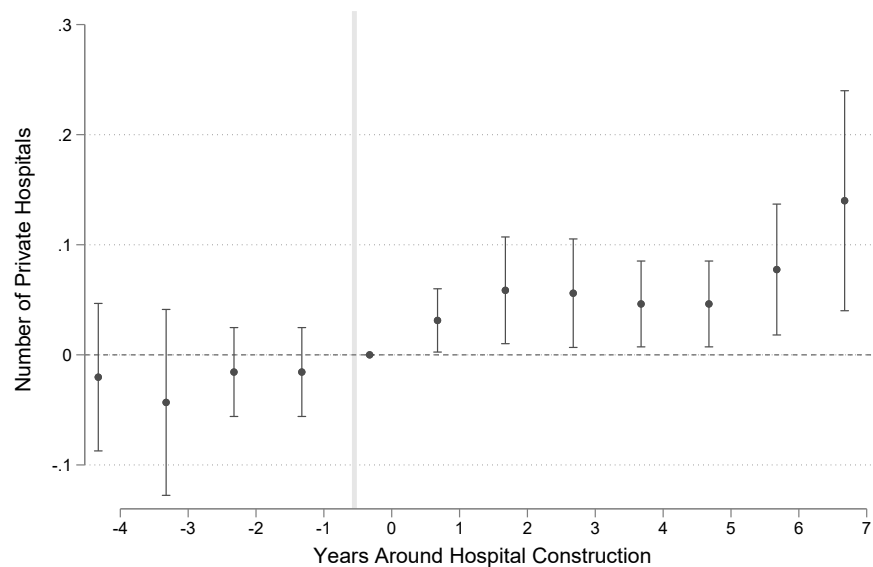


Note: Panel A: all public hospitals. Panel B: specialist public hospitals. Panel C: non-specialist public hospitals. Control units selected using the Synthetic DiD procedure. Shaded area represents the 95 percent confidence interval.

existing public hospitals in 1996, which is a policy-relevant baseline characteristic given the Ministry of Health's stated objective of building in districts without existing public capacity. The matching procedure reduces the sample from 47 districts (25 treatment districts, 22 control) to 28 districts (12 treated, 16 control).

Table B.13 shows that matching removes all statistically significant differences. Figure B.6 presents the event study for the matched sample, pooling specialist and non-specialist public hospitals. The average post-treatment effect is 0.108 additional private hospitals, representing a 31 percent increase relative to the matched sample's pre-treatment mean. The attenuated magnitude relative to the specialist-only estimate (0.785) is expected, since the matched sample includes non-specialist events that reduce the pooled average.

Figure B.6: Event Study of Matching on Private Hospital Entrants



Note: Event study estimates using the Sun and Abraham (2021) estimator on the coarsened exact matched sample (12 treated, 16 control districts). Combines both specialist and non-specialist public hospitals. Matching variables: rurality and number of existing public hospitals in 1996. The average post-treatment effect is 0.108. Standard errors clustered at the district level.

Table B.13: Post-Matching Summary Statistics by Treatment Status

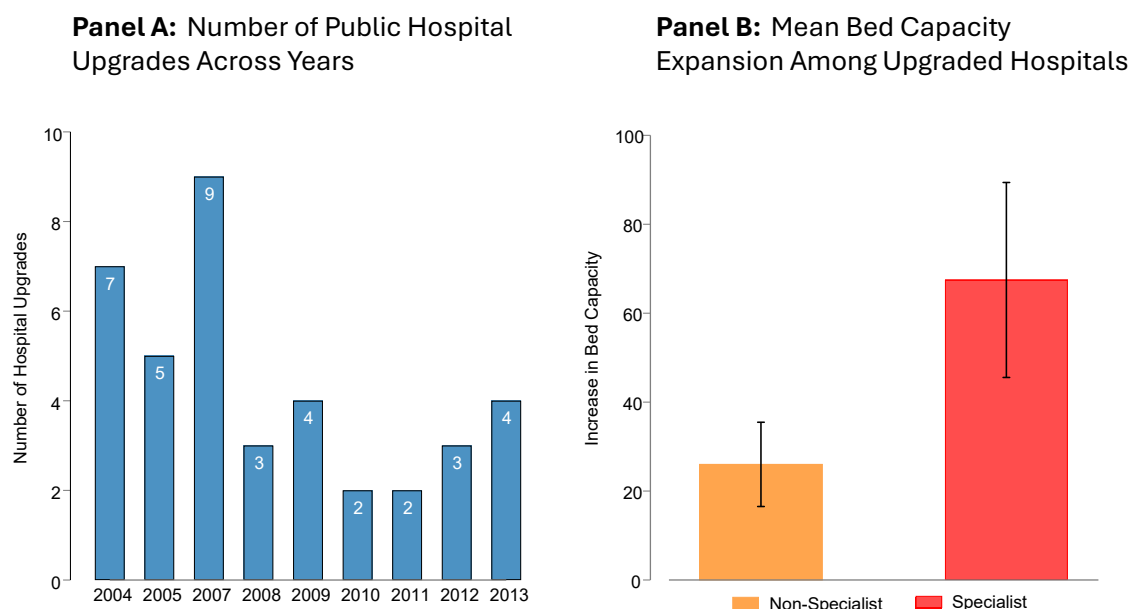
Variable	Treated (N=12)	Never Treated (N=16)	Difference	P-value
<i>Panel A. Demographics</i>				
Log Population	11.186 (0.870)	11.001 (0.869)	0.185	0.582
Rural Population Share	0.843 (0.224)	0.850 (0.262)	-0.007	0.936
Chinese Share	0.127 (0.116)	0.170 (0.198)	-0.043	0.505
Malay Share	0.402 (0.362)	0.431 (0.384)	-0.029	0.839
Indian Share	0.061 (0.077)	0.039 (0.063)	0.022	0.429
Married Share	0.375 (0.015)	0.398 (0.041)	-0.023	0.078
Financial Services Employment Share	0.005 (0.015)	0.006 (0.011)	-0.001	0.894
<i>Panel B. Education</i>				
College/University Education	0.012 (0.014)	0.014 (0.017)	-0.002	0.772
Secondary Education Completed	0.170 (0.077)	0.205 (0.064)	-0.035	0.200
Primary Education Completed	0.188 (0.048)	0.191 (0.038)	-0.003	0.859
Some Primary Education	0.216 (0.028)	0.209 (0.031)	0.007	0.550
<i>Panel C. Age Distribution</i>				
Age <1	0.031 (0.008)	0.026 (0.008)	0.005	0.106
Age 1–4	0.119 (0.019)	0.110 (0.020)	0.009	0.233
Age 5–18	0.340 (0.039)	0.314 (0.044)	0.026	0.119
Age 19–45	0.369 (0.063)	0.388 (0.062)	-0.019	0.451
Age 46–60	0.095 (0.031)	0.101 (0.031)	-0.006	0.612
Age 61–74	0.034 (0.017)	0.047 (0.020)	-0.013	0.077
Age >74	0.012 (0.009)	0.015 (0.008)	-0.003	0.347
<i>Panel D. Health Facilities</i>				
Number of Private Hospitals	0.094 (0.401)	0.375 (1.500)	-0.281	0.534
Number of Public Hospitals	0.125 (0.345)	0.125 (0.342)	0.000	1.000
Number of Private Doctors	12.500 (36.927)	18.750 (62.915)	-6.250	0.762
Distance to Nearest Public Hospital (km)	33.054 (13.585)	36.570 (40.181)	-3.516	0.774
Distance to Nearest Private Hospital (km)	118.454 (88.809)	113.839 (122.953)	4.615	0.913

Notes: Standard deviations in parentheses. Difference = Treated - Never Treated. P-values from two-sample t-tests. Sample reduced from 47 districts to 28 districts (12 treated, 16 control) after coarsened exact matching on rurality and number of existing public hospitals.

B.11 Falsification: Hospital Upgrades

To further test the physician training mechanism, I examine an alternative treatment: public hospital upgrades that expand bed capacity without creating new training infrastructure. Unlike new construction, which creates training programs, residency positions, and specialist faculty, upgrades add beds to facilities where teaching programs already exist. If crowd-in operates through new training capacity rather than general service expansion, upgrades should produce muted effects.

Figure B.7: Number of Public Hospital Upgrades by Year



Note: Hospital upgrades data are from the Ministry of Health's annual Health Facts publications, which report bed capacity at each public hospital. Upgrades are defined as year-over-year increases in bed capacity at existing public hospitals, 2003–2012.

Between 2003 and 2012, 35 public hospitals underwent significant bed capacity expansions (Figure B.7). Table B.14 supports the prediction: specialist upgrades produce an effect less than one-third the size of new specialist construction, and the estimate is not distinguishable from zero. Non-specialist upgrades show a negative point estimate similar to new non-specialist construction. The contrast between new construction and upgrades is consistent with the physician training

mechanism: creating entirely new training infrastructure generates large physician supply spillovers, while marginally expanding existing capacity does not.

Table B.14: Effects of Public Hospital Upgrades on Private Hospital Entry

	Number of Private Hospitals		
	(1)	(2)	(3)
All upgrades	0.138 (0.173)		
Specialist upgrades		0.238 (0.215)	
Non-specialist upgrades			−0.170 (0.169)
Mean Dep. Var.	0.934	1.132	0.631
Pre-Treatment Mean	1.028	1.480	0.000
District Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	1026	846	594
R^2	0.947	0.946	0.928
Events	35	24	11
Estimator	SA	SA	SA

Notes: Each column presents results from separate regressions examining the impact of public hospital upgrades (2003–2012) using the Sun and Abraham (2021) estimator. Hospital upgrades are defined as year-over-year increases in bed capacity, identified from the Ministry of Health’s annual Health Facts publications. Control units are districts that never received upgrades. Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Standard errors clustered at the district level in parentheses.

B.12 Heterogeneous Effects by Private Hospital Size

The physician supply mechanism predicts that crowd-in effects should concentrate among smaller hospitals, which require fewer specialists to reach minimum viable scale. [Table B.15](#) supports this prediction. Specialist hospitals generate large crowd-in effects on small private hospitals (fewer than 94 beds), increasing their count by roughly half relative to the pre-treatment stock, but show essentially no effect on large hospitals (Column 2). This is consistent with small hospitals benefiting

most from marginal increases in the local specialist pool, while large hospitals in urban areas face relatively abundant physician supply regardless of public hospital construction. Non-specialist hospitals show directionally negative effects for both size categories (Columns 3–4), consistent with demand competition operating independently of hospital scale. Event study dynamics are in [Figure B.3](#).

Table B.15: Effects on Private Hospitals by Size

	Specialist		Non-Specialist	
	Small (1)	Large (2)	Small (3)	Large (4)
Post $\times$ Treated	0.727 (0.085)	0.057 (0.029)	−0.173 (0.005)	−0.059 (0.002)
Mean Dep. Var.	0.910	0.790	0.268	0.364
Pre-Treatment Mean	1.429	1.143	0.000	0.000
Observations	648	648	594	594
R^2	0.911	0.982	0.851	0.980
Events	14	14	11	11
District Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Estimator	SA	SA	SA	SA

Notes: Each column presents results from separate regressions using the Sun and Abraham (2021) estimator. The dependent variable is the number of private hospitals in each size category. Small: fewer than 94 beds; large: 94 or more beds. Pre-Treatment Mean is the mean outcome among treated districts in the pre-treatment period. Standard errors clustered at the district level in parentheses.

B.13 Further Details on Implied Capacity Calculation

The alternative placement in [Section 5.4](#) ranks the 84 candidate districts (no in-window specialist treatment, no pre-1996 specialist hospital) by predicted crowd-in from the entry function. Predicted crowd-in at district d is $\hat{\tau}_d = \hat{N}_d (e^{\beta\delta} - 1)$, truncated at zero, where $\hat{N}_d$ is the fitted count from equation (14) at 1996 population with the recalibrated intercept, and $\delta = 0.124$ is the profit shift from [Section 5.4](#). The construction anchors to the ATT: predicted crowd-in summed across the 14 treated districts equals the estimated total by choice of δ .

Panel A of [Table B.16](#) reports both sides of the reallocation: the six zero-baseline treated districts whose placements are removed, and the top six candidates that receive them. The removed districts hold no private hospitals at baseline, and their fitted counts imply predicted crowd-in below 0.05 hospitals each. Panel B aggregates. The net gain of 3.51 crowded-in private hospitals equals the additions at the six new sites (3.64) less the forgone crowd-in at the vacated sites (0.13). Beds convert at the 1996–2013 entrant mean of 55 per private hospital.

Table B.16: Implied Private Capacity Under Observed and Alternative Specialist Placement

Panel A: Reallocated placements and predicted crowded-in effects			
Removed		Added (top-ranked candidates)	
District	Predicted effect	District	Predicted effect
Temerloh	0.05	Petaling	2.44
Bintulu	0.14	Timur Laut	1.33
Batang Padang	0.23	Larut dan Matang	1.09
Sarikei	0.00	Pasir Mas	0.19
Lahad Datu	0.00	Rompin	0.00
Sepang	0.00	Mersing	0.00
Sum	0.42	Sum	5.05
Panel B: Aggregate implied capacity			
	Observed	Alternative	Difference
Crowded-in private hospitals	14.0	18.6	4.6
Crowded-in private beds	770	1,025	255

Notes: Predicted effects apply Column (2) of [Table B.18](#) to 1996 log population, truncated at zero. Candidate ranking and the accounting of the alternative column are described in the text. Sums may not match differences exactly due to rounding.

B.14 Effects on Other Health Workforce Categories

Table B.17 asks whether the labor spillover extends beyond specialist physicians to the rest of the workforce the public sector trains. As with physicians, I measure the private-practice margin by self-employment, since the census does not record employment sector directly and public-sector training is common to all these professions. Specialist hospitals raise the total headcount of nurses and health technicians at the new facility (Columns 1–2), but neither profession shows any increase in self-employment (Columns 3–4).

An important caveat to mention is that self-employment is a good proxy for private specialist physicians but not private nurses or other health worker categories. Nurses and health technicians are salaried in both sectors and cannot establish independent practices, and public-sector nurses in Malaysia earn slightly more than their private-sector counterparts, removing even the wage incentive to switch. The null on self-employed nurses and technicians therefore rules out the possibility in which a public hospital pushes every trained profession into self-employment, and suggests that the labor channel operates through the specialist physician pipeline rather than a general public-sector workforce expansion.

B.15 Heterogeneous Effects by Baseline District Characteristics

Where does the crowd-in effect occur across treated districts? In the framework, a specialist hospital raises variable profit through the specialist pool, but every entry response scales with population and disappears below the entry threshold (Prediction 4.2). Crowd-in should therefore concentrate in districts near the private entry threshold and with access to a specialist pool, and should be absent where demand is too thin for private entry.

Table B.17: Effects of Public Hospitals on Nurses and Health Technicians (100s)

	Total		Self-Employed	
	Specialist (1)	Non-spec. (2)	Specialist (3)	Non-spec. (4)
<i>Panel A. Nurses and Midwives (100s)</i>				
Post $\times$ Treated	4.373 (2.377)	−0.640 (0.574)	0.102 (0.109)	−0.033 (0.034)
Mean Dep. Var.	4.654	2.360	0.092	0.061
Pre-Treatment Mean	6.944	0.333	0.056	0.000
Observations	120	114	120	114
R^2	0.919	0.962	0.775	0.844
<i>Panel B. Health Technicians (100s)</i>				
Post $\times$ Treated	10.593 (5.493)	−1.747 (1.451)	0.288 (0.237)	−0.179 (0.102)
Mean Dep. Var.	5.254	2.943	0.196	0.145
Pre-Treatment Mean	3.611	0.333	0.000	0.000
Observations	120	114	120	114
R^2	0.748	0.854	0.760	0.841
District $\times$ Stack FE	Yes	Yes	Yes	Yes
Year $\times$ Stack FE	Yes	Yes	Yes	Yes

Notes: Stacked difference-in-differences estimates using census data (1970, 1980, 1991). Dependent variables are in hundreds. Nurses and midwives identified by ISCO-68 codes 71 and 73 (ISCO-88 code 323). Health technicians identified by ISCO-68 code 77 (ISCO-88 codes 322–323). Columns 3–4 count self-employed workers, the proxy used for private practice throughout; self-employment is a defensible proxy for private practice among specialists but not among nurses and technicians, who are salaried in both sectors. Specialist hospitals raise total nursing and technical headcount (Columns 1–2) but show no increase in self-employment (Columns 3–4). Standard errors clustered by district in parentheses.

I test this prediction by interacting my treatment variable with four separate pre-treatment characteristics that capture distinct channels. First, a binary indicator for a private hospital in 1996 measures viability—the market has already proven it can sustain private entry. Log population in 1996 measures latent demand and how close a market sits to the entry threshold. The log stock of public-sector doctors measures the depth of the local physician pool that a private entrant can recruit from. A binary indicator for a public hospital in 1996 captures whether private hospitals simply cluster around public health infrastructure regardless of whether it trains physicians. I estimate two-way fixed-effects specifications on the full panel of districts between 1996 and 2013:

$$Y_{dt} = \mu_d + \lambda_t + \beta_1 \text{Spec}_{dt} + \beta_2 (\text{Spec}_{dt} \times X_d) + \gamma \text{NonSpec}_{dt} + u_{dt} \quad (15)$$

where Spec_{dt} and NonSpec_{dt} indicate post-treatment years for specialist and non-specialist construction, and X_d is one predetermined characteristic, centered at the sample mean for the continuous measures. β_1 measures the specialist effect for a district at the mean. Column 5 includes all four interactions jointly. The non-specialist indicator enters as a control in every column so that non-specialist-treated districts do not serve as untreated comparisons during their own post-treatment years. [Table B.18](#) shows that every characteristic predicts a larger specialist effect in isolation (Columns 1–4). However, when stacking all four interactions to separate these channels, log of population explains the crowd-in (Column 5). The log population interaction, at approximately 1.307, is essentially unchanged. The doctor interaction falls to approximately 0.162 but is marginally statistically significant. This is consistent with a residual labor-market margin beyond what population absorbs. The public hospital interaction collapses from approximately

1.834 to essentially zero. The crowd-in therefore scales with district population, as Prediction 4.2 implies.

One big caveat is that the interactions are identified from 14 specialist-treated clusters. Standard errors at this cluster count likely understate uncertainty. I therefore read Table B.18 as descriptive support for Prediction 4.2.

Table B.18: Heterogeneity of Specialist Hospital Effect by Pre-treatment District Characteristics

	Number of Private Hospitals				
	(1)	(2)	(3)	(4)	(5)
Specialist public hospitals	−0.189 (0.094)	−0.309 (0.245)	0.311 (0.236)	0.037 (0.200)	−0.265 (0.254)
× Private Hospital in 1996	1.662 (0.547)				−0.564 (0.490)
× Log Population (1996, centered)		1.287 (0.266)			1.307 (0.280)
× Log Public Doctors (1991, centered)			0.354 (0.121)		0.162 (0.084)
× Public Hospital in 1996				1.834 (0.690)	0.148 (0.459)
Non-specialist public hospitals	−0.251 (0.070)	−0.251 (0.070)	−0.251 (0.070)	−0.253 (0.070)	−0.251 (0.070)
Mean Dep. Var.	0.738	0.738	0.738	0.738	0.738
District Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
N Districts × Year	2,340	2,340	2,340	2,340	2,340
R^2	0.954	0.957	0.955	0.955	0.958

Notes: Each column reports a TWFE regression of the number of private hospitals on separate post-treatment indicators for specialist and non-specialist construction, with the specialist indicator interacted with one predetermined characteristic; Column (5) includes all four interactions. The sample excludes three districts with missing 1991 doctor counts. Standard errors clustered at the district level in parentheses.

B.16 Additional Tables and Figures

Table B.19: Effects of Specialist Public Hospitals on Private Entry by Distance Band

	Number of Private Hospitals	
	Post-treatment average	Pre-treatment mean
0–2.5km	–0.028 (0.059)	0.286
2.5–5km	–0.084 (0.050)	0.286
5–10km	0.014 (0.060)	1.429
10–15km	0.187 (0.098)	0.214
Beyond 15km	–0.117 (0.051)	0.357
District Fixed Effects	Yes	
Year Fixed Effects	Yes	
Treated Districts	14	
Estimator	SA	

Notes: Each row reports the average post-treatment coefficient from a separate estimation of [Equation 13](#), tabulating [Figure 6](#) Panel B. The dependent variable is the number of private hospitals within the stated band of the new specialist public hospital in treated districts, and the number of private hospitals in the entire district in never-treated districts. Every band regression uses the same comparison group, so each coefficient carries the same never-treated trend; that common term cancels when bands are differenced, and the coefficients are informative about the gradient across bands rather than the level of entry in any one band. Pre-treatment means are measured among treated districts in the year before opening. Standard errors clustered at the district level in parentheses.

Table B.20: Distance Gradient With and Without District Clipping

	Band Differential in Private Hospitals	
	Rings across districts (1)	Rings within district (2)
0–2.5km	–0.061 (0.087)	0.029 (0.053)
2.5–5km	–0.108 (0.067)	–0.018 (0.018)
5–10km (ref.)	—	—
10–15km	0.359 (0.233)	0.350 (0.182)
Outer band	0.072 (0.109)	0.000 (0.046)
Events	14	14
N event $\times$ Year $\times$ Band	810	810

Notes: Band differentials relative to the 5–10km reference band and the year before opening, from regressions of private hospital counts on band-by-pre and band-by-post indicators with district-by-year and district-by-band fixed effects, on the 14 specialist public hospitals in single-hospital districts over event years –4 to 7. Identification is within event and year, with no comparison districts, so the coefficients are differentials across bands of the same hospital rather than the band-level estimates of [Table B.19](#). Column (1) draws rings across district boundaries and assigns each private hospital within 25km to its nearest new specialist hospital; hospitals beyond 25km of any event (48 of 134) are excluded and the outer band is 15–25km. Column (2) confines rings to the hospital’s own district, with the outer band extending from 15km to the district boundary. The trend-adjusted contrast nets the pre-treatment differential. Standard errors clustered at the district level in parentheses (14 clusters).

Table B.21: Entry Function: Poisson Regression and Diagnostics

	Number of Private Hospitals		
	(1)	(2)	(3)
Log population	2.433 (0.303)	1.462 (0.489)	1.890 (0.367)
Log(1 + public doctors)		0.394 (0.199)	
Census-Year Fixed Effects	Yes	Yes	Yes
Districts	133	130	47
N District $\times$ Census Year	399	278	141
Pseudo R^2	0.591	0.600	0.570

Notes: Poisson regressions of district private hospital counts on log population, estimated on the census panel (1970, 1980, and 1991) with census-year fixed effects, equation (14). Column (1) is the primary specification on all 133 districts. Column (2) adds the log of one plus the district's public doctor stock; three districts drop for missing 1991 doctor data. Column (3) re-estimates column (1) on the 47 districts of the event-study estimation sample. Standard errors clustered at the district level in parentheses.